

2022 阿里巴巴全球数学竞赛

单选题：磁性几何魔方

把实心立方体 $ABCD - A_1B_1C_1D_1$ (假设 $AB = 1$) 按如下方式分成 12 块 (图1) :

- 1) 取 6 条面上的对角线 $AC, AB_1, AD_1, C_1B, C_1D, C_1A_1$;
- 2) 考虑以立方体中心为顶点, 上述 6 条对角线及 12 条棱之一为对边的三角形;
- 3) 这 18 个三角形把立方体切成了 12 块, 每块是一个四面体, 每个四面体有两条棱是立方体的棱;
- 4) 每个四面体仅通过其上立方体的棱和其它四面体连结.

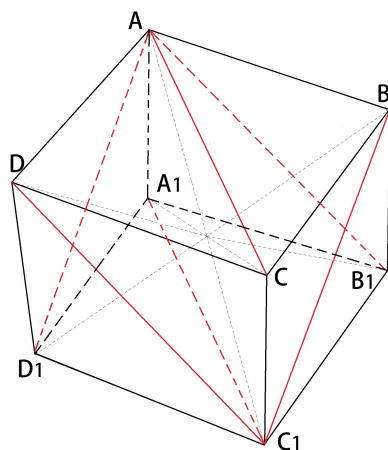


图1：磁性几何魔方

这样一个玩具可以摆出各种形状 (图2) .

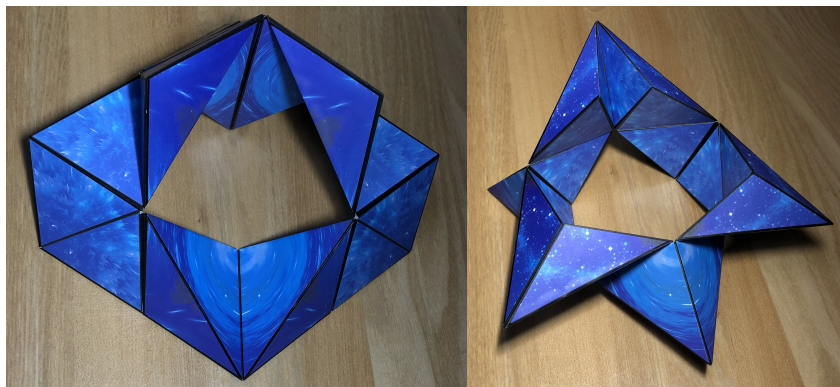


图2：其它形状的例子

问：在这个玩具所有可能的形状中,其上两点之间(在空间中)距离的最大值是多少?

$\sqrt{11}$ $\sqrt{7+4\sqrt{2}}$ $\sqrt{13}$ $1+2\sqrt{2}$ 以上都不对

单选题：间隔围观

某日，某地的一片巨大空场地上有一场街头艺术表演，引起部分群众围观。我们将此地看作一个欧氏平面，表演的区域中心记作点 K 。

设群众为 $A_1, A_2, \dots, A_n, \dots$ ，他们按照从 A_1 开始的顺序依次选定一个围观的位置

$$P_1, P_2, \dots, P_n, \dots,$$

但需同时满足以下三个条件。

条件1: A_n 围观的位置与 K 的距离不小于 10 米，即对任意 n , $KP_n \geq 10$ 米；

条件2: A_n 围观的位置与前面每个人位置的距离都需要不小于 1 米，即对任意 $n \geq 2$ 和任意 $1 \leq m \leq n-1$, $P_m P_n \geq 1$ 米。

条件3: 在满足条件1 和条件2 的前提下， A_n 选择与 K 尽量接近的点进行围观，即他希望 KP_n 取到最小可能的值。如果同时满足条件1 和条件2 且使得 KP_n 取到最小可能值的 P_n 不止一个，那么 A_n 可以选择其中的任意一个点。

例如，对于 A_1 ，他选择位置时没有条件2，故他会选择以 K 为圆心，10 米为半径的圆周 C 上的任意一点（与 K 的距离恰为 10 米）；对于 A_2 ，他也希望 P_2 与 K 的距离恰为 10 米，即 P_2 也在 C 上。由于 C 上有许多点与 P_1 的距离不小于 1 米，他可以选择其中任意一点。

(1) 请问，以下说法哪个正确？

存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 \leq KP_n \leq c_2$ （单位：米）；

存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 \sqrt{n} \leq KP_n \leq c_2 \sqrt{n}$ （单位：米）；

存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 n \leq KP_n \leq c_2 n$ （单位：米）；

存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 n^2 \leq KP_n \leq c_2 n^2$ （单位：米）。

(2) 由于人是有一定体积的，所以如果某人站在另一人围观的视线路径附近时，第二人就会被第一人挡住视线。我们认为，对不同的 i, j ，如果以 P_i 为圆心， $\frac{1}{6}$ 米为半径的圆周与线段 KP_j 相交，那么 A_j 就会被 A_i 挡住视线而看不到表演的全貌。

请问，以下哪个说法正确？

当有 60 名群众围观时，一定有人看不到表演的全貌；

当有 60 名群众围观时，存在所有人都看到表演全貌的可能性，
但当有 800 名群众围观时，一定有人看不到表演的全貌；

当有 800 名群众围观时，存在所有人都看到表演全貌的可能性，
但当有 10000 名群众围观时，一定有人看不到表演的全貌；

当有 10000 名群众围观时，存在所有人都看到表演全貌的可能性。

单选题：“虎虎生威”盲盒

春节期间，“勇敢牛牛”牛奶公司推出了新春集福活动：每盒牛奶都附赠一个红包，红包中藏有下列“虎”，“生”，“威”中的一款图案。（如下图）



集齐两个“虎”，一个“生”，一个“威”即可拼齐成为“虎虎生威”全家福。这项活动一经推出，就成为了网红爆款，很多人希望能够集齐一整套。假设红包中的图案是独立随机分布的（并且不能从红包外观上进行区别），“虎”，“生”，“威”三款红包按均匀概率 $\frac{1}{3}$ 分布。

(1) 收集齐一整套“虎虎生威”全家福所需要购买的牛奶盒数的数学期望是多少？

$6\frac{1}{3}$;

$7\frac{1}{3}$;

$8\frac{1}{3}$;

$9\frac{1}{3}$;

以上都不对。

(2) 在市场部的周会讨论中，大家认为当前的图案投放比例，会导致在收集“虎虎生威”全家福时收集到太多的“生”和“威”，于是探讨可能的改进方案。记图案“虎”、“生”和“威”的投放比例为 (p, q, r) ，那么下面哪种方案下，收集齐一整套“虎虎生威”全家福所需要购买的牛奶盒数的数学期望是最小的？

$(p, q, r) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$;

$(p, q, r) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$;

$(p, q, r) = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$;

$(p, q, r) = (\frac{3}{4}, \frac{1}{8}, \frac{1}{8})$.

证明题

给定集合 X , 若函数 $f : X \times X \rightarrow [0, 1]$ 满足: 对任意 $\epsilon > 0$, 存在有限个 X 中元素 $b_1, b_2, \dots, b_m$ 使得对任意 $t \in X$, 存在某个 $b_{i(t)}$ 满足:

$$|f(x, t) - f(x, b_{i(t)})| < \epsilon, \forall x \in X,$$

则称 f 是右一致的。类似地, 对任意 $\epsilon > 0$, 存在有限个 X 中元素 $a_1, a_2, \dots, a_n$ 使得对任意 $t \in X$, 存在某个 $a_{i(t)}$ 满足:

$$|f(t, x) - f(a_{i(t)}, x)| < \epsilon, \forall x \in X,$$

则称 f 是左一致的。求证: 左一致等价于右一致。

证明题

设 n 是正整数, $V = \mathbb{R}^n$ 是 n 维欧氏空间, 有一组基 $e_i = (\underbrace{0, \dots, 0}_{i-1}, 1, \underbrace{0, \dots, 0}_{n-i})$ ($1 \leq i \leq n$) 满足

$$(e_i, e_j) = \delta_{i,j},$$

其中

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

是Kronecker 符号, $(\cdot, \cdot)$ 是 V 上的内积. 对 V 中的非零向量 v , 定义线性变换 $s_v : V \rightarrow V$ 为

$$s_v(u) = u - \frac{2(u, v)}{(v, v)}v, \quad \forall u \in V.$$

对于介于 0 和 n 之间的整数 k , 记 $\text{Gr}_k(V)$ 为 V 的 k 维子空间的集合. 对于 V 的一个 k 维子空间 W , 记 $[W]$ 为 $\text{Gr}_k(V)$ 中的相应元素. 取 W 的一组正交规范基 $\{v_1, \dots, v_k\}$ (也即, $(v_i, v_j) = \delta_{i,j}$), 定义 $s_{[W]} : V \rightarrow V$ 为

$$s_{[W]} = s_{v_1} \cdots s_{v_k}.$$

(1) 证明 $s_{[W]}$ 不依赖于正交规范基 $\{v_1, \dots, v_k\}$ 的选取.

(2) 证明 $s_{[W]}^2 = \text{id}$.

(3) 对 $[W'] \in \text{Gr}_k(V)$, 定义

$$t_{[W]}([W']) = [s_{[W]}(W')],$$

其中 $s_{[W]}(W')$ 是 W' 在 $s_{[W]}$ 下的像. 我们称 $\text{Gr}_k(V)$ 的子集 X 为一个“有趣集”若

$$t_{[W]}([W']) = [W'], \quad \forall [W], [W'] \in X.$$

请找到 $\text{Gr}_k(V)$ 中有趣集的最大的元素个数, 并证明之.

解答题：忙碌的快递员

一位快递员在二维格点平面上坐标为 $(n, 0)$ 处取到了快递。他所工作的快递站则在原点 $(0, 0)$ 处。此后，快递员每一步做步长为 1 的简单随机游动。在以下的各问中，不妨认为正整数 n 远大于 1。

- (1) 令 $P_{1,n}$ 为该快递员在走了恰好 $\lfloor n^{1.5} \rfloor$ 步时，距离快递站的直线距离大于 $\frac{n}{2}$ 的概率，试证明

$$\lim_{n \rightarrow +\infty} P_{1,n} = 1$$

- (2) 令 $P_{2,n}$ 为该快递员在前 $\lfloor n^{1.5} \rfloor$ 步内曾经回到过快递站的概率，试证明

$$\lim_{n \rightarrow +\infty} P_{2,n} = 0$$

- (3) 令 $P_{3,n}$ 为该快递员的前 2^n 步内曾经回到过快递站的概率，试证明

$$\lim_{n \rightarrow +\infty} P_{3,n} = 1$$

解答题

- (1) 证明不存在满足如下条件的周期序列 $a_1, a_2, a_3, \dots$, 每项均为 ± 1 , 且对任意有理数 θ ,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

- (2) 证明不存在每项均为 ± 1 的序列 $a_1, a_2, a_3, \dots$ 满足: 对任意有理数 θ

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq 2022.$$

- (3) 举例说明: 存在每项均为 ± 1 的序列 $a_1, a_2, a_3, \dots$ 满足: 其中有无穷多项 1, 也有无穷多项 -1 , 且对任意 $\theta \in \mathbb{Q} - \mathbb{Z}$,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

解答题：开幕式的节目设计

假设你被冬奥会开幕式的导演选为技术助理，负责用数学知识研究设计方案的合理性。在开幕式的备选节目中，有一个方案是让一群由无人机控制的吉祥物绕一个圆圈形状的场地滑冰。因为无人机足够的多（但是不会拥堵或者相撞），我们认为可以用一个概率密度函数 $\rho(t, v) (\geq 0)$ 来刻画无人机的分布。因为场地是圆环形的，所以我们可以认为速度 $v \in \mathbb{R}$ ，它表示无人机的线速度。那么，对于任意给定的时间 t ，和两个速度 $v_1 < v_2$ ，

$$\int_{v_1}^{v_2} \rho(t, v) dv$$

表示全体无人机中速度介于 v_1 和 v_2 之间的概率。

由于无人机的运动机理，已知这个密度函数的演化满足如下的方程

$$\rho_t + ((u(t) - v)\rho)_v = \rho_{vv}, \quad v \in \mathbb{R}, \quad t > 0.$$

其中 $u(t)$ 为无人机的指令速度。

(1) 为了研究怎么给无人机合适的指令速度，大宝给导演提议说，应该让

$$u(t) = u_0 + u_1 N(t)$$

其中 $u_0 > 0$ ， $u_1 > 0$ ，而 $N(t)$ 表示无人机的速度正部 ($v_+ = \max\{0, v\}$) 的平均

$$N(t) = \int_{-\infty}^{+\infty} v_+ \rho(t, v) dv = \int_0^{+\infty} v \rho(t, v) dv.$$

但是，你善意地提醒道，如果 $u_1 > 1$ 那么 $N(t)$ 在演化过程中不会有上界，以至于很快引起无人机的故障。你可以证明你的上述结论吗？（为了方便讨论，我们忽略 ρ 及其导数在 $|v| \rightarrow +\infty$ 时的贡献。）

(2) 在采纳了大宝和你的建议后，导演又在考虑这些无人机是否会在滑行中在圆圈场地上均匀分布，于是我们需要考虑关于速度和位置的联合密度函数 $p(t, x, v) (\geq 0)$ 。这里 $x \in [0, 2\pi]$ 表示无人机在圆圈上的相对位置，显然 $\int_0^{2\pi} p(t, x, v) dx = \rho(t, v)$ 。已知这个联合密度函数的演化满足

$$p_t + vp_x + ((u(t) - v)p)_v = p_{vv}, \quad x \in [0, 2\pi], \quad v \in \mathbb{R}, \quad t > 0.$$

且由于无人机在绕圈滑行，在 x 方向上如下条件满足

$$p(t, 0, v) = p(t, 2\pi, v), \quad v \in \mathbb{R}, \quad t > 0.$$

你大胆猜测：无论初始分布如何，无人机很快就在圆圈上接近一个均匀分布。你可以证明或者证伪这个命题吗？（为了方便讨论，我们忽略 p 及其导数在 $|v| \rightarrow +\infty$ 时的贡献。）

2022 阿里巴巴全球数学竞赛（预选赛）

第1题（单选题）磁性几何魔方

把实心立方体 $ABCD - A_1B_1C_1D_1$ (假设 $AB = 1$) 按如下方式分成 12 块（图1）：

- 1) 取 6 条面上的对角线 $AC, AB_1, AD_1, C_1B, C_1D, C_1A_1$;
- 2) 考虑以立方体中心为顶点, 上述 6 条对角线及 12 条棱之一为对边的三角形;
- 3) 这 18 个三角形把立方体切成了 12 块, 每块是一个四面体, 每个四面体有两条棱是立方体的棱;
- 4) 每个四面体仅通过其上立方体的棱和其它四面体连结.

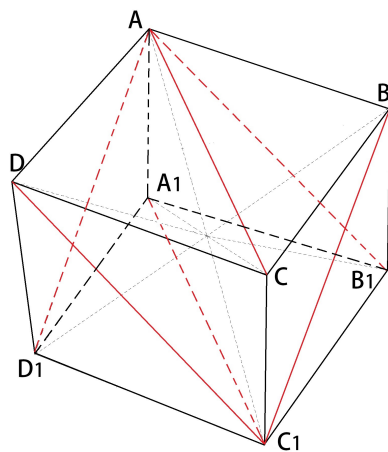


图1: 磁性几何魔方

这样一个玩具可以摆出各种形状（图2）.

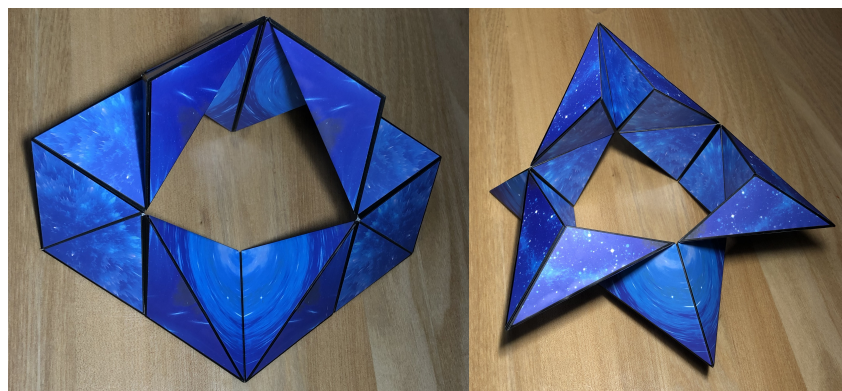


图2: 其它形状的例子

问: 在这个玩具所有可能的形状中,其上两点之间(在空间中)距离的最大值是多少?

(A) $\sqrt{11}$ (B) $\sqrt{7+4\sqrt{2}}$ (C) $\sqrt{13}$ (D) $1+2\sqrt{2}$ (E) 以上都不对

1 答案 选D。以 $ABCD-A_1B_1C_1D_1$ 标记立方体的八个顶点, 并设开始时选定的对顶点为 A 和 C_1 . 那么, 这两个顶点分别属于6个小四面体, 而 B, C, D, D_1, A_1, B_1 这六个顶点(无论怎么变形都)构成一个边长为1的六边形(可能退化).

注意到这个玩具由12个四面体(从而是凸图形)构成, 因此取到最大值的必定是两个小四面体的顶点.

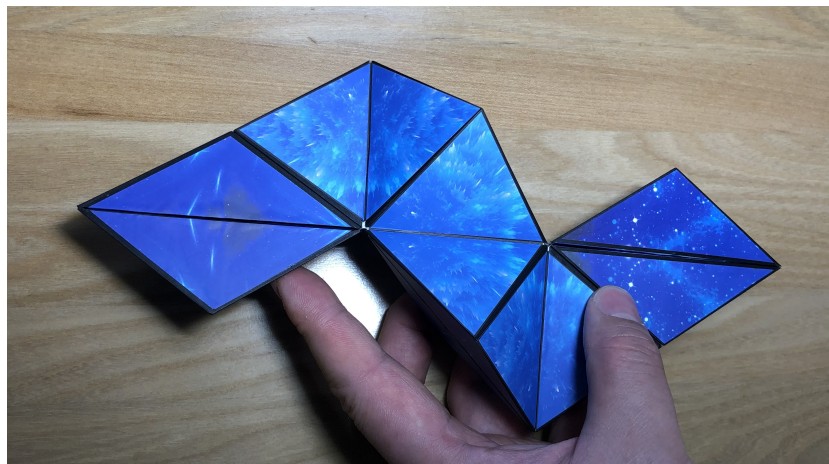
在一个四面体中, 四个顶点可分为3类:

- 一个顶点在立方体中对应于 A 或 C_1 , 我们称之为**第一类顶点**;
- 两个顶点(从而它们之间的棱)属于(空间)六边形 $BCDD_1A_1B_1$, 我们称之为**第二类顶点**;
- 一个顶点对应于立方体的中心, 它到同一个小四面体的另三个顶点的距离都是 $\frac{\sqrt{3}}{2}$, 我们称之为**第三类顶点**.

我们考察这些顶点之间**沿棱**的距离的最大值, 易得下表:

距离	第一类顶点	第二类顶点	第三类顶点
第一类顶点	$1+2\sqrt{2}$	$2+\sqrt{2}$	$1+\sqrt{2}+\frac{\sqrt{3}}{2}$
第二类顶点	$2+\sqrt{2}$	3	$2+\frac{\sqrt{3}}{2}$
第三类顶点	$1+\sqrt{2}+\frac{\sqrt{3}}{2}$	$2+\frac{\sqrt{3}}{2}$	$2+\sqrt{3}$

从而这个玩具上任意两点之间的距离不会超过 $1+2\sqrt{2}$. 而这个值是可以达到的, 如下图所示:



第2,3题（单选题）间隔围观

某日，某地的一片巨大空场地上有一场街头艺术表演，引起部分群众围观。我们将此地看作一个欧氏平面，表演的区域中心记作点 K 。

设群众为 $A_1, A_2, \dots, A_n, \dots$ ，他们按照从 A_1 开始的顺序依次选定一个围观的位置

$$P_1, P_2, \dots, P_n, \dots,$$

但需同时满足以下三个条件。

条件1: A_n 围观的位置与 K 的距离不小于 10 米，即对任意 n , $KP_n \geq 10$ 米；

条件2: A_n 围观的位置与前面每个人位置的距离都需要不小于 1 米，即对任意 $n \geq 2$ 和任意 $1 \leq m \leq n-1$, $P_m P_n \geq 1$ 米。

条件3: 在满足条件1 和条件2 的前提下， A_n 选择与 K 尽量接近的点进行围观，即他希望 KP_n 取到最小可能的值。如果同时满足条件1 和条件2 且使得 KP_n 取到最小可能值的 P_n 不止一个，那么 A_n 可以选择其中的任意一个点。

例如，对于 A_1 ，他选择位置时没有条件2，故他会选择以 K 为圆心，10 米为半径的圆周 C 上的任意一点（与 K 的距离恰为 10 米）；对于 A_2 ，他也希望 P_2 与 K 的距离恰为 10 米，即 P_2 也在 C 上。由于 C 上有许多点与 P_1 的距离不小于 1 米，他可以选择其中任意一点。

(1) 请问，以下说法哪个正确？

- (A) 存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 \leq KP_n \leq c_2$ （单位：米）；
- (B) 存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 \sqrt{n} \leq KP_n \leq c_2 \sqrt{n}$ （单位：米）；
- (C) 存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 n \leq KP_n \leq c_2 n$ （单位：米）；
- (D) 存在正实数 c_1, c_2 ，使得对任意正整数 n ，无论 $A_1, A_2, \dots, A_n$ ，怎么选择位置，均有 $c_1 n^2 \leq KP_n \leq c_2 n^2$ （单位：米）。

(2) 由于人是有一定体积的，所以如果某人站在另一人围观的视线路径附近时，第二人就会被第一人挡住视线。我们认为，对不同的 i, j ，如果以 P_i 为圆心， $\frac{1}{6}$ 米为半径的圆周与线段 KP_j 相交，那么 A_j 就会被 A_i 挡住视线而看不到表演的全貌。

请问，以下哪个说法正确？

- (A) 当有 60 名群众围观时，一定有人看不到表演的全貌；
- (B) 当有 60 名群众围观时，存在所有人都看到表演全貌的可能性，但当有 800 名群众围观时，一定有人看不到表演的全貌；
- (C) 当有 800 名群众围观时，存在所有人都看到表演全貌的可能性，但当有 10000 名群众围观时，一定有人看不到表演的全貌；
- (D) 当有 10000 名群众围观时，存在所有人都看到表演全貌的可能性。

2 答案 选B。设 KP_n 的长度为 d_n 米. 一方面, 我们以 $P_1, P_2, \dots, P_{n-1}$ 为圆心, 1 米为半径作圆, 由 P_n 的选取可知这些圆再加上 C 的内部必然覆盖了以 K 为圆心, d_n 为半径的圆, 因此

$$\pi \cdot d_n^2 \leq (n-1) \cdot \pi \cdot 1^2 + \pi \cdot 10^2,$$

故

$$d_n \leq \sqrt{n+99} \leq \sqrt{100n} = 10\sqrt{n}.$$

另一方面, 我们以 $P_1, P_2, \dots, P_n$ 为圆心, $\frac{1}{2}$ 米为半径作圆, 由于 $P_1, P_2, \dots, P_n$ 两两之间距离不小于 1 米, 故这些圆彼此不相交. 另外, 由题意可知 $KP_1, KP_2, \dots, KP_n$ 均应不超过 d_n 米 (若有 $1 \leq m \leq n-1$ 使得 KP_m 的长度超过了 d_n 米, 则 A_m 可以选择距离更近的 P_n 点, 矛盾), 因此, 这些圆都在以 K 为圆心, $d_n + \frac{1}{2}$ 米为半径的圆内, 因此

$$\pi(d_n + \frac{1}{2})^2 \geq n \cdot \pi \cdot (\frac{1}{2})^2,$$

故

$$d_n \geq \frac{\sqrt{n}}{2} - \frac{1}{2}.$$

注意当 $n=1$ 时, $d_1=10$, 而对 $n \geq 2$, $\frac{1}{2} < \frac{2\sqrt{n}}{5}$, 故

$$d_n > \frac{\sqrt{n}}{2} - \frac{2\sqrt{n}}{5} = \frac{\sqrt{n}}{10}.$$

因此, $\frac{\sqrt{n}}{10} \leq d_n \leq 10\sqrt{n}$, B 选项正确.

3 答案 选B。一方面, 当有 60 人围观时, 由于 $\sin \frac{\pi}{60} > \frac{1}{10} \sin \frac{\pi}{6} = \frac{1}{20}$, 故 $20 \sin \frac{\pi}{60} > 1$, 这说明一个内接于 C 的正 60 边形的边长不小于 1 米. 因此, 当 $P_1, P_2, \dots, P_{60}$ 恰好为这个正 60 边形的所有顶点时, 符合题目条件. 而对任意不同的 i, j , 点 P_i 到 KP_j 的距离不小于 $10 \sin \frac{\pi}{30}$ 米, 由 $\sin \frac{\pi}{30} > \frac{1}{5} \sin \frac{\pi}{6} = \frac{1}{10}$ 可知这些距离都不小于 1 米, 故所有人都能看到表演的全貌.

另一方面, 当有 800 人围观时, 作射线 $\overrightarrow{KP_1}, \overrightarrow{KP_2}, \dots, \overrightarrow{KP_{800}}$, 如果有两条射线重合 (不妨设为 $\overrightarrow{KP_i}, \overrightarrow{KP_j}$ 重合), 设 $KP_j > KP_i$, 那么 A_j 被 A_i 挡住了视线.

假设没有重合的射线, 我们先证明, 如果存在一个角 $\angle P_i KP_j$ 满足 $\angle P_i KP_j \leq \frac{1}{12} \left(\frac{1}{KP_i} + \frac{1}{KP_j} \right)$ (角的单位为弧度, 线段的单位为米), 那么, A_i, A_j 中有一人被另一人挡住了视线.

不失一般性, 我们假设 $KP_i \leq KP_j$, 由 $KP_i, KP_j \geq 10$ 知 $\angle P_i KP_j \leq \frac{1}{60}$, 故它是锐角. 因此 P_i 到 KP_j 的垂足位于线段 KP_j 的内部, 而 P_i 到 KP_j 的距离为

$$KP_i \sin \angle P_i KP_j < KP_i \cdot \angle P_i KP_j \leq \frac{KP_i}{12} \left(\frac{1}{KP_i} + \frac{1}{KP_j} \right) \leq \frac{1}{6},$$

故 A_j 被 A_i 挡住了视线.

考虑射线 $\overrightarrow{KP_1}, \overrightarrow{KP_2}, \dots, \overrightarrow{KP_{800}}$ 把以 K 为顶点的周角分成的 800 个角，它们的和为 2π ，但是每个角对应的 $\frac{1}{12} \left(\frac{1}{KP_i} + \frac{1}{KP_j} \right)$ 的和

$$\begin{aligned}
 \sum \frac{1}{12} \left(\frac{1}{KP_i} + \frac{1}{KP_j} \right) &= \frac{1}{6} \sum_{i=1}^{800} \frac{1}{KP_i} \\
 &\geq \frac{1}{6} \sum_{i=1}^{800} \frac{1}{\sqrt{i+99}} \quad (\text{第(1)问的结论}) \\
 &= \frac{1}{6} \sum_{m=100}^{899} \frac{1}{\sqrt{m}} \\
 &\geq \frac{1}{6} \sum_{m=100}^{899} \left(\int_m^{m+1} \frac{1}{\sqrt{x}} dx \right) \quad (\text{因为 } \frac{1}{\sqrt{x}} \text{ 是减函数}) \\
 &= \frac{1}{6} \int_{100}^{900} \frac{1}{\sqrt{x}} dx = \frac{\sqrt{900} - \sqrt{100}}{3} = \frac{20}{3} > 2\pi.
 \end{aligned}$$

因此，存在一个角 $\angle P_i K P_j$ 满足 $\angle P_i K P_j \leq \frac{1}{12} \left(\frac{1}{KP_i} + \frac{1}{KP_j} \right)$ 故必然有人看不到表演的全貌.
综上所述，B 选项正确.

第4,5题（单选题）“虎虎生威”盲盒

春节期间，“勇敢牛牛”牛奶公司推出了新春集福活动：每盒牛奶都附赠一个红包，红包中藏有下列“虎”，“生”，“威”中的一款图案。（如下图）



集齐两个“虎”，一个“生”，一个“威”即可拼齐成为“虎虎生威”全家福。这项活动一经推出，就成为了网红爆款，很多人希望能够集齐一整套。假设红包中的图案是独立随机分布的（并且不能从红包外观上进行区别），“虎”，“生”，“威”三款红包按均匀概率 $\frac{1}{3}$ 分布。

(1) 收集齐一整套“虎虎生威”全家福所需要购买的牛奶盒数的数学期望是多少？

- (A) $6\frac{1}{3}$;
- (B) $7\frac{1}{3}$;
- (C) $8\frac{1}{3}$;
- (D) $9\frac{1}{3}$;
- (E) 以上都不对。

(2) 在市场部的周会讨论中，大家认为当前的图案投放比例，会导致在收集“虎虎生威”全家福时收集到太多的“生”和“威”，于是探讨可能的改进方案。记图案“虎”、“生”和“威”的投放比例为 (p, q, r) ，那么下面哪种方案下，收集齐一整套“虎虎生威”全家福所需要购买的牛奶盒数的数学期望是最小的？

- (A) $(p, q, r) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$;
- (B) $(p, q, r) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$;
- (C) $(p, q, r) = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$;
- (D) $(p, q, r) = (\frac{3}{4}, \frac{1}{8}, \frac{1}{8})$.

4 答案 选B。可以通过连续时间嵌入到Poisson过程的解法得到一般的解析表达式。假设盲盒总计有 n 个图案，每次收集到图案 i 的概率是 p_i ，以及最终目标是对于每个图案 i 需要收集 k_i 个。记首次达成全收集时购买的盲盒数量是 N ，目标即计算 $\mathbb{E}[N]$ 。我们将这个过程嵌入到连续时间的Poisson过程中：假设有一个参数为1的Poisson点过程，每次信号到达时，都独立的按照概率 p_i 抽样图案。记停时

$$T_i = \inf\{t \in \mathbb{R}_+ : \text{在时间} t \text{ 收集到了} k_i \text{ 个图案} i\}$$

$$T = \max_{1 \leq i \leq n} T_i.$$

事实上有如下结论： $\mathbb{E}[T] = \mathbb{E}[N]$ 。我们先证明这个结论。可以将Poisson点过程到达的第 j 个信号所需的时间记作 τ_j ，那么有

$$T = \sum_{j=1}^N \tau_j.$$

事实上 $(\tau_j)_{j \geq 1}$ 是独立同分布的，参数为1的指数分布。利用条件概率，我们有

$$\begin{aligned} \mathbb{E}[T] &= \mathbb{E}\left[\sum_{j=1}^N \tau_j\right] \\ &= \sum_{k=0}^{\infty} \mathbb{E}\left[\sum_{j=1}^k \tau_j \mid N = k\right] \mathbb{P}[N = k] \\ &= \sum_{k=0}^{\infty} \mathbb{P}[N = k] k \mathbb{E}[\tau_1] \\ &= \mathbb{E}[N]. \end{aligned}$$

接下来我们只需要计算 $\mathbb{E}[T]$ 。首先我们使用Fubini定理

$$\begin{aligned} \mathbb{E}[T] &= \int_0^{\infty} \mathbb{P}[T > t] dt \\ &= \int_0^{\infty} (1 - \mathbb{P}[T \leq t]) dt \\ &= \int_0^{\infty} (1 - \mathbb{P}[T_i \leq t, \forall 1 \leq i \leq n]) dt. \end{aligned}$$

事实上，连续时间的Poisson过程的thinning性质为我们创造了很多独立性，使得计算变得简单。比如此处我们可以把收集每一个图案 i 看作是参数为 p_i 的独立的 n 个Poisson过程。于是有

$$\mathbb{E}[T] = \int_0^{\infty} \left(1 - \prod_{i=1}^n \mathbb{P}[T_i \leq t]\right) dt.$$

同时， $\mathbb{P}[T_i \leq t]$ 可以直接利用Poisson分布写出表达式

$$\mathbb{P}[T_i \leq t] = 1 - \sum_{k=0}^{k_i-1} e^{-p_i t} \frac{(p_i t)^k}{k!}.$$

我们可以得到最终的表达式

$$\mathbb{E}[N] = \int_0^\infty \left(1 - \prod_{i=1}^n \left(1 - \sum_{k=0}^{k_i-1} e^{-p_i t} \frac{(p_i t)^k}{k!} \right) \right) dt.$$

在我们的题目中， $n = 3$ ，目标收集 $(k_1, k_2, k_3) = (2, 1, 1)$ 。我们可以计算得到解析的表达式

$$\mathbb{E}[N] = 1 + p_1 + \left(\frac{2}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} \right) - \sum_{i=1}^3 \frac{1}{1-p_i} - \frac{p_1}{(p_1+p_2)^2} - \frac{p_1}{(p_1+p_3)^2}.$$

在 $(p_1, p_2, p_3) = (1/3, 1/3, 1/3)$ 得到期望为 $7\frac{1}{3}$ ，答案选B。

5 答案 选C。上一问中计算得到方案A的期望是 $7\frac{1}{3}$ ，D显然不是一个好的方案，因为收集到“威”字图案的数学期望是8，已经超出了方案A。所以核心在于考察方案B，C。方案B在这里非常符合直观，是一个带有暗示的陷阱。利用解法2中得到的公式

$$\begin{aligned} \mathbb{E}[\tau] = 1 + p + \left(\frac{2}{p} + \frac{1}{q} + \frac{1}{r} \right) - \left(\frac{1}{p+q} + \frac{1}{p+r} + \frac{1}{q+r} \right) \\ - \frac{p}{(p+q)^2} - \frac{p}{(p+r)^2}. \end{aligned}$$

B,C对应的期望分别为 $7\frac{1}{18}, 6\frac{223}{245}$ 。所以事实上方案C是最佳的，选C。

第6题（证明题）

给定集合 X , 若函数 $f : X \times X \rightarrow [0, 1]$ 满足: 对任意 $\epsilon > 0$, 存在有限个 X 中元素 $b_1, b_2, \dots, b_m$ 使得对任意 $t \in X$, 存在某个 $b_{i(t)}$ 满足:

$$|f(x, t) - f(x, b_{i(t)})| < \epsilon, \forall x \in X,$$

则称 f 是右一致的。类似地, 对任意 $\epsilon > 0$, 存在有限个 X 中元素 $a_1, a_2, \dots, a_n$ 使得对任意 $t \in X$, 存在某个 $a_{i(t)}$ 满足:

$$|f(t, x) - f(a_{i(t)}, x)| < \epsilon, \forall x \in X,$$

则称 f 是左一致的。求证: 左一致等价于右一致。

6 答案 只需证明右一致推出左一致。任给 $\epsilon > 0$, 由右一致条件得到 X 中有限个元素 $b_1, b_2, \dots, b_m$. 考虑映射 $h : X \rightarrow [0, 1]^m, h(x) = (f(x, b_1), f(x, b_2), \dots, f(x, b_m))$. 由 $[0, 1]^m$ 的紧致性, 存在有限个 $c_1, c_2, \dots, c_n$ 使得对任意 $x \in X$, 存在某个 c_{i_x} 使得 $|h(x) - h(c_{i_x})| < \epsilon$, 从而

$$|f(x, b_i) - f(c_{i_x}, b_i)| < \epsilon, \forall i \in \{1, 2, \dots, m\}. \quad (1)$$

由右一致条件知 $|f(x, t) - f(x, b_{i(t)})| < \epsilon (\forall x \in X), |f(c_{i_x}, t) - f(c_{i_x}, b_{i(t)})| < \epsilon$, 结合(1), 知对任意 $t \in X$,

$$|f(x, t) - f(c_{i_x}, t)| \leq |f(x, t) - f(x, b_{i(t)})| + |f(x, b_{i(t)}) - f(c_{i_x}, b_{i(t)})| + |f(c_{i_x}, b_{i(t)}) - f(c_{i_x}, t)| < 3\epsilon.$$

因为 ϵ 是任意的, 所以 f 是左一致的。

第7题（证明题）

设 n 是正整数, $V = \mathbb{R}^n$ 是 n 维欧氏空间, 有一组基 $e_i = (\underbrace{0, \dots, 0}_{i-1}, 1, \underbrace{0, \dots, 0}_{n-i})$ ($1 \leq i \leq n$) 满足

$$(e_i, e_j) = \delta_{i,j},$$

其中

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

是Kronecker 符号, $(\cdot, \cdot)$ 是 V 上的内积. 对 V 中的非零向量 v , 定义线性变换 $s_v: V \rightarrow V$ 为

$$s_v(u) = u - \frac{2(u, v)}{(v, v)}v, \quad \forall u \in V.$$

对于介于 0 和 n 之间的整数 k , 记 $\text{Gr}_k(V)$ 为 V 的 k 维子空间的集合. 对于 V 的一个 k 维子空间 W , 记 $[W]$ 为 $\text{Gr}_k(V)$ 中的相应元素. 取 W 的一组正交规范基 $\{v_1, \dots, v_k\}$ (也即, $(v_i, v_j) = \delta_{i,j}$), 定义 $s_{[W]}: V \rightarrow V$ 为

$$s_{[W]} = s_{v_1} \cdots s_{v_k}.$$

(1) 证明 $s_{[W]}$ 不依赖于正交规范基 $\{v_1, \dots, v_k\}$ 的选取.

(2) 证明 $s_{[W]}^2 = \text{id}$.

(3) 对 $[W'] \in \text{Gr}_k(V)$, 定义

$$t_{[W]}([W']) = [s_{[W]}(W')],$$

其中 $s_{[W]}(W')$ 是 W' 在 $s_{[W]}$ 下的像. 我们称 $\text{Gr}_k(V)$ 的子集 X 为一个“有趣集”若

$$t_{[W]}([W']) = [W'], \quad \forall [W], [W'] \in X.$$

请找到 $\text{Gr}_k(V)$ 中有趣集的最大的元素个数, 并证明之.

7 答案 (1) 记 $W^\perp$ 为 W 在 V 中的正交补. 则

$$s_{[W]}|_W = -1 \text{ 且 } s_{[W]}|_{W^\perp} = 1.$$

由此我们刻画了 $s_{[W]}$ 并证明了它与 W 的正交规范基 $\{v_1, \dots, v_k\}$ 的选取无关.

(2) 由于

$$s_{[W]}|_W = -1 \text{ 且 } s_{[W]}|_{W^\perp} = 1,$$

所以 $s_{[W]}^2 = \text{id}$.

(3) (i) 我们证明

$$t_{[W]}([W']) = [W'] \Leftrightarrow W' = (W' \cap W) \oplus (W' \cap W^\perp).$$

充分性易证. 必要性: 假设 $t_{[W]}([W']) = [W']$. 则对任意的 $u \in W'$, 我们有 $s_{[W]}(u) \in W'$. 记 $u = u_1 + u_2$, 其中 $u_1 \in W$ 且 $u_2 \in W^\perp$. 则 $-u_1 + u_2 = s_{[W]}(u) \in W'$. 所以, $u_1 \in W'$ 且 $u_2 \in W'$. 因此,

$$W' \subset (W' \cap W) \oplus (W' \cap W^\perp).$$

显然, $(W' \cap W) \oplus (W' \cap W^\perp) \subset W'$. 这样,

$$W' = (W' \cap W) \oplus (W' \cap W^\perp).$$

(ii) 记

$$X = \{\text{span}_{\mathbb{R}}\{e_{i_1}, \dots, e_{i_k}\} : 1 \leq i_1 < \dots < i_k \leq n\}.$$

由(i) 中的判别准则, X 是个“有趣集”. 这个集合 X 的元素个数为 $\binom{n}{k}$.

(iii) 对 n 作归纳, 我们证明: $\text{Gr}_k(V)$ 中有趣集 X 的元素个数不超过 $\binom{n}{k}$. 当 $n = 1$ 时, 这是显然的. 假设此结论在 $n < m$ 时成立. 设 $n = m$. 当 $k = 0$ 或者 m 时, 结论显然成立. 设 $1 \leq k \leq m - 1$. 取 $[W] \in X$. 对每个整数 i ($0 \leq i \leq k$), 记

$$X_i = \{[W'] \in X : \dim W' \cap W = k - i\}.$$

对每个 $[W'] \in X_i$, 记

$$Y_i = \{[W' \cap W] : [W'] \in X_i\} \subset \text{Gr}_{k-i}(W)$$

和

$$Z_i = \{[W' \cap W^\perp] : [W'] \in X_i\} \subset \text{Gr}_i(W^\perp).$$

由(i) 中的判别准则, Y_i 是 $\text{Gr}_{k-i}(W)$ 中的有趣集且 Z_i 是 $\text{Gr}_i(W^\perp)$ 中的有趣集. 由于 $\dim W = k < m$ 且 $\dim W^\perp = m - k < m$, 由归纳假设我们得到

$$|Y_i| \leq \binom{k}{k-i} \text{ 且 } |Z_i| \leq \binom{m-k}{i}.$$

这样,

$$|X| = \sum_{0 \leq i \leq k} |X_i| \leq \sum_{0 \leq i \leq k} |Y_i| |Z_i| \leq \sum_{0 \leq i \leq k} \binom{k}{k-i} \binom{m-k}{i} = \binom{m}{k}.$$

由此证明了结论.

第8题（解答题）忙碌的快递员

一位快递员在二维格点平面上坐标为 $(n, 0)$ 处取到了快递。此后，快递员每一步做步长为 1 的简单随机游动。他所工作的快递站则在原点 $(0, 0)$ 处。在以下的各问中，不妨认为正整数 n 远大于 1。

- (1) 令 $P_{1,n}$ 为该快递员在走了恰好 $\lfloor n^{1.5} \rfloor$ 步时，距离快递站的直线距离大于 $\frac{n}{2}$ 的概率，试证明

$$\lim_{n \rightarrow +\infty} P_{1,n} = 1$$

- (2) 令 $P_{2,n}$ 为该快递员在前 $\lfloor n^{1.5} \rfloor$ 步内曾经回到过快递站的概率，试证明

$$\lim_{n \rightarrow +\infty} P_{2,n} = 0$$

- (3) 令 $P_{3,n}$ 为该快递员的前 2^n 步内曾经回到过快递站的概率，试证明

$$\lim_{n \rightarrow +\infty} P_{3,n} = 1$$

8 答案 (1) 考虑该快递员在走了恰好 $\lfloor n^{1.5} \rfloor$ 步时距离快递站的直线距离大概率小于等于 $\frac{1}{2}$ 的事件。由三角不等式我们有这一事件必然包含于以下事件 A

$$A = \text{快递员此时距取件点的距离大于等于 } \frac{1}{2}$$

进一步根据三角不等式，如果事件 A 发生，则快递员在至少一个坐标方向上移动的距离至少为 $\frac{n}{4}$ 。不妨设该方向为 y -轴方向。该快递员第 k 步时在 y -轴方向的移动距离可以被看做是以下独立同分布的随机变量的求和：

$$Y_k = X_1 + X_2 + \cdots + X_k$$

其中 $P(X_i = 0) = \frac{1}{2}, P(X_i = \pm 1) = \frac{1}{4}$ 。易见。这样 $E(X_i) = 0, \text{var}(X_i) = \frac{1}{2}$ ，由切比雪夫不等式，我们有

$$P(A) \leq 2P\left(\left|Y_{\lfloor n^{1.5} \rfloor}\right| \geq \frac{n}{4}\right) \leq 2 \times \frac{n^{1.5}}{2n^2/16} = 16n^{-0.5} < 1$$

(2) 与上一问相同，由三角不等式我们有这一事件必然包含于以下事件 A

$$A = \text{快递员在前 } \lfloor n^{1.5} \rfloor \text{ 步内到达过距取件点的距离大于等于 } n \text{ 的地方}$$

如果事件 A 发生，则快递员在至少一个坐标方向上一定至少移动过至少 $\frac{n}{2}$ 的距离。即

$$P(A) \leq 2P\left(\max_{k \leq n^{1.5}} |Y_k| \geq \frac{n}{2}\right)$$

注意到 Y_k 是一个鞅，根据Doob 不等式

$$P\left(\max_{k \leq n^{1.5}} |Y_k| \geq \frac{n}{2}\right) \leq \frac{E\left[\left|Y_{\lfloor n^{1.5} \rfloor}\right|^2\right]}{(n/2)^2} \leq \frac{n^{1.5}}{2(n/2)^2} = 2n^{-0.5} < 1$$

(3) 令 $S_k \in \mathbb{Z}^2, k \geq 0$ 为快递员在第 k 步时的位置。我们定义以下停时

$$\tau = \min\{k \geq 0, S_k = 0\}$$

$$\hat{\tau} = \min\{k \geq 0, |S_k| \geq 2^{n/4}\}$$

因此，对于我们关心的概率，有

$$P(\tau > 2^n) \leq P_{(n,0)}(\hat{\tau} > 2^n) + P_{(n,0)}(\hat{\tau} < \tau)$$

其中 $P_{(n,0)}$ 为从 $(n,0)$ 点出发的简单随机游动的轨道概率分布。对于上式中的第一项，由不变原理，我们有 $S_{\lfloor 2^{n/2}t \rfloor} 2^{-\frac{n}{4}}$ 在 $t \in [0, 1]$ 上收敛到 0 出发的二维标准布朗运动 B_t 。对于布朗运动易见

$$P(|B_1| \geq 2) = c > 0$$

对于一切充分大的 n ，结合平移不变性我们有

$$\min_{|X_0| \leq 2^{n/4}} P_{X_0}(|S_{2^{n/2}}| \geq 2^{n/4}) \geq P_0(|S_{2^{n/2}}| \geq 2 \times 2^{n/4}) \geq P(|B_1| \geq 2) - \frac{c}{2} \geq \frac{c}{2}$$

下面我们对快递员的 2^n 步按 $2^{n/2}$ 的长度分段。如果该快递员在 2^n 步中始终位于距离 $(0,0)$ 点 $2^{n/4}$ 的半径之内，则他必须在步数为 $2^{n/2}, 2 \times 2^{n/2}, \dots, [2^{n/2}]2^{n/2}$ 时始终距离 $(0,0)$ 点 $2^{n/4}$ 的半径之内。再由随机游动的马氏性，我们有

$$P_{(n,0)}(\hat{\tau} > 2^n) \leq \left[1 - \min_{|X_0| \leq 2^{n/4}} P_{X_0}(|S_{2^{n/2}}| \geq 2^{n/4})\right]^{[2^{n/2}]} \leq \left(1 - \frac{c}{2}\right)^{[2^{n/2}]} \ll 1$$

而对于第二项 $p_2 = P_{(n,0)}(\hat{\tau} < \tau)$ 这里我们需要考虑二维简单随机游动的位势核(potential kernel)，对于任意 $x \in \mathbb{Z}^2$ 考虑

$$a(x) = \sum_{k=0}^{+\infty} [P_0(S_k = 0) - P_0(S_k = x)]$$

由Kesten (1987, Lemma 1)¹ 可知

- 1) $a(x)$ 中的级数求和对任意 $x \in \mathbb{Z}^2$ 收敛且非负
- 2) 对于任意初始位置出发的简单随机游动 S_k , $a(S_{k \wedge \tau})$ 是一个非负鞅
- 3) 存在常数 C_0 使得

$$|a(x) - \log |x| / (2\pi) - C_0| = O(|x|^{-2})$$

结合上述3条以及鞅的停时分解定理，我们有

$$a((n,0)) = (1 - p_2) \times 0 + p_2 E[a(S_{\hat{\tau}}) | \tau > \hat{\tau}]$$

因此

$$p_2 = \frac{a((n,0))}{E[a(S_{\hat{\tau}}) | \tau > \hat{\tau}]}$$

注意到由上述第3条：

$$a((n,0)) \asymp \log(n)$$

$$E[a(S_{\hat{\tau}}) | \tau > \hat{\tau}] \asymp n$$

因此， $p_2 \ll 1$ 证毕。

¹H. Kesten, Hitting probabilities of random walks on $\mathbb{Z}^d$, Stochastic Process. Appl. 25 (1987) 165 - 184.

第9题（解答题）

- (1) 证明不存在满足如下条件的周期序列 $a_1, a_2, a_3, \dots$, 每项均为 ± 1 , 且对任意有理数 θ ,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

- (2) 证明不存在每项均为 ± 1 的序列 $a_1, a_2, a_3, \dots$ 满足: 对任意有理数 θ

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq 2022.$$

- (3) 举例说明: 存在每项均为 ± 1 的序列 $a_1, a_2, a_3, \dots$ 满足: 其中有无穷多项 1, 也有无穷多项 -1 , 且对任意 $\theta \in \mathbb{Q} - \mathbb{Z}$,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

9 答案 (1) 简记 $z := e^{2\pi i\theta}$. 用反证法, 假设存在周期为 $d \in \mathbb{N}$ 的由 ± 1 构成的序列 (a_n) 满足

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n z^n \right| < +\infty,$$

其中 $z = e^{2\pi i\theta}$, $\theta \in \mathbb{Q}$.

取 N 为 d 的倍数, $N = dM$. 因为 (a_n) 是周期的, 我们可以分组求和:

$$\begin{aligned} \sum_{n=1}^{Md} a_n z^n &= (a_1 z + a_2 z^2 + \dots a_d z^d) + z^d (a_1 z + a_2 z^2 + \dots a_d z^d) \\ &\quad + \dots + z^{(M-1)d} (a_1 z + a_2 z^2 + \dots a_d z^d) \\ &= (a_1 z + a_2 z^2 + \dots a_d z^d) (1 + z^d + z^{2d} + \dots + z^{(M-1)d}). \end{aligned}$$

取 $\theta = k/d \in \mathbb{Q}$, 其中 $k \in \{0, \dots, d-1\}$ 待定. 因为 $z^d = (e^{2\pi i k/d})^d = 1$, 所以

$$\sum_{n=1}^{Md} a_n z^n = M(a_1 z + a_2 z^2 + \dots a_d z^d).$$

只需取 k 使得 $a_1 z + a_2 z^2 + \dots a_d z^d$ 非零就得到了矛盾:

$$\lim_{M \rightarrow +\infty} \left| \sum_{n=1}^{Md} a_n z^n \right| = \lim_{M \rightarrow +\infty} M |a_1 z + a_2 z^2 + \dots a_d z^d| = +\infty.$$

而若 $a_1 z + a_2 z^2 + \dots a_d z^d$ 对 $k \in \{0, \dots, d-1\}$ (其中 $z = e^{2\pi i k/d}$) 均为 0, 则多项式 $P(X) = a_1 X + a_2 X^2 + \dots a_d X^d$ 有 $d+1$ 个不同复根:

$$e^{2\pi i 0/d}, e^{2\pi i 1/d}, \dots, e^{2\pi i (d-1)/d} \quad \text{以及} \quad 0,$$

这是不可能的。

(2) 同样用反证法, 假设 (a_n) 为这样的序列. 首先, 因为

$$f_N := \sum_{n=1}^N a_n e^{2\pi i n \theta}$$

是 θ 的连续函数, 对所有实数也有 $|f_N(\theta)| \leq 2022$.

我们计算 f_N 的 L^2 模. 注意到对 $n \in \mathbb{Z}$,

$$\int_0^1 e^{2\pi i n \theta} d\theta = \begin{cases} 1 & \text{若 } n = 0, \\ 0 & \text{若 } n \neq 0. \end{cases}$$

从而我们有

$$\begin{aligned}
\int_0^1 |f_N|^2 d\theta &= \int_0^1 \sum_{n=1}^N a_n e^{2\pi i n \theta} \sum_{m=1}^N a_m e^{-2\pi i m \theta} d\theta \\
&= \sum_{n=1}^N \sum_{m=1}^N a_n a_m \underbrace{\int_0^1 e^{2\pi i (n-m)\theta} d\theta}_{\text{非零当且仅当 } n=m} \\
&= \sum_{n=1}^N a_n^2 = N
\end{aligned}$$

由此就能看出

$$\left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq 2022$$

不能对所有的 N 和 $\theta \in \mathbb{R}$ 成立。否则的话就有 $\int_0^1 |f_N|^2 d\theta \leq 2022^2$, 与 $\int_0^1 |f_N|^2 d\theta = N$ ($N > 2022^2$), 矛盾!

(3) (虽然最终常值的序列满足条件, 但是并不符合我们的要求。)

为了说明想法的由来, 我们先看看 $\theta = 1/3$ 以及 $N = 3M$ 的情况。我们要找 a_i 使得

$$\sup_{M \in \mathbb{N}} \left| \left(\sum_{n=0}^{M-1} a_{3n+1} \right) \cdot 1 + \left(\sum_{n=0}^{M-1} a_{3n+2} \right) \cdot e^{2\pi i/3} + \left(\sum_{n=0}^{M-1} a_{3n+3} \right) \cdot e^{4\pi i/3} \right| < +\infty$$

这对应于伸缩三个向量 $1, e^{2\pi i/3}$ 和 $e^{4\pi i/3}$, 然后计算和的模长。因为 $1 + e^{2\pi i/3} + e^{4\pi i/3} = 0$, 我

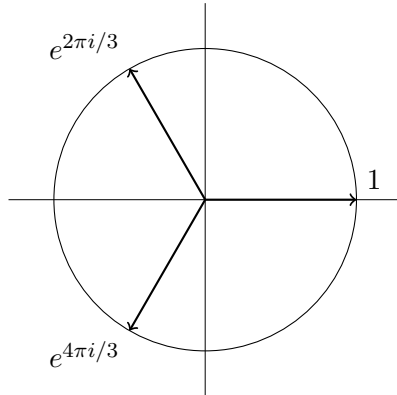


Figure 3: $\theta = 1/3$ 的情况.

们设法让这三个向量具有相同的模长让它们尽量抵消。换言之, 对 $i = 1, i = 2$ 以及 $i = 3$, 序列 $(a_{3n+i})_{n \geq 1}$ 中 1 和 -1 的分布是差不多的。类似于这里的 3 , 这样的性质应该对所有大于 1 的正整数成立。

有很多中方法构造这样的 (a_n) , 一种可行的方法如下:

$$\begin{array}{c|c|c|c|c|c|c|c|c|c|c|c|c} n & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \hline a_n & +1 & -1 & -1 & +1 & +1 & +1 & +1 & +1 & +1 & -1 & -1 \end{array},$$

其中+1出现1!次, 接着-1出现2!次, 然后+1出现3! = 6次, 如此继续下去。下面证明该序列满足要求。

对 $k \geq 0$, 令 $f(k) = 0! + 1! + \cdots + k!$. 用 $I_k \subset \mathbb{N}$ 表示满足 $f(k) \leq n < f(k+1)$ 的自然数 n . 从而当 $n \in I_k$ 时, $a_n = (-1)^k$. 该序列的关键特征是当 $q \geq 2$ 时, I_k 的长度是 $(k+1)!$, 从而当 k 充分大时可被 q 整除。

设 $\theta = p/q$ 是有理数, 其中 $\gcd(p, q) = 1, q > 1$. 将题中的三角和写成如下形式:

$$\sum_{n=1}^N a_n e^{2\pi i n \theta} = \sum_{k \geq 0} (-1)^k \sum_{\substack{n \in I_k \\ n \leq N}} e^{2\pi i n p/q}.$$

下面给出该和不依赖于 N 的界。

考虑其中一段, $S_k := \sum_{\substack{n \in I_k \\ n \leq N}} e^{2\pi i n p/q} (k \geq 0)$.

(i) 若 $k < q$, 由绝对值不等式 $|S_k| \leq f(k+1) - f(k) = (k+1)!$

(ii) 若 $k \geq q$, 我们利用序列的构造证明除了最后一段以外均为0.

(a) 若 $k \geq q$ 且 $f(k+1) \leq N$, 则 $S_k = 0$. 因为根据等比级数求和

$$S_k = \sum_{n=f(k)}^{f(k+1)-1} e^{2\pi i n p/q} = \frac{(e^{2\pi i p/q})^{f(k+1)} - (e^{2\pi i p/q})^{f(k)}}{e^{2\pi i p/q} - 1}.$$

因为 $k \geq q$, $(k+1)!$ 能被 q 整除, 分子为0:

$$\left(e^{2\pi i p/q}\right)^{f(k)+(k+1)!} - \left(e^{2\pi i p/q}\right)^{f(k)} = \left(e^{2\pi i p/q}\right)^{f(k)} \left(e^{2\pi i p(k+1)!/q} - 1\right) = 0.$$

(b) 若 $k \geq q$ 而 $N \in I_k$ (即最后一段), 我们有

$$S_k = \sum_{n=f(k)}^N e^{2\pi i n p/q} = \frac{(e^{2\pi i p/q})^{N+1} - (e^{2\pi i p/q})^{f(k)}}{e^{2\pi i p/q} - 1}.$$

由绝对值不等式有

$$|S_k| \leq \frac{\left| (e^{2\pi i p/q})^N \right| + \left| (e^{2\pi i p/q})^{f(k)} \right|}{\left| e^{2\pi i p/q} - 1 \right|} = \frac{2}{\left| e^{2\pi i p/q} - 1 \right|}.$$

(c) 若 $k \geq q$ 且 $N < f(k)$, 则 $S_k = 0$, 实际上该段为空。

因此

$$\left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq \sum_{k=0}^{q-1} |S_k| + \sum_{k \geq q} |S_k| \leq f(q) + \frac{2}{|e^{2\pi i p/q} - 1|}.$$

即

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

第10题（解答题）开幕式的节目设计

假设你被冬奥会开幕式的导演选为技术助理，负责用数学知识研究设计方案的合理性。在开幕式的备选节目中，有一个方案是让一群由无人机控制的吉祥物绕一个圆圈形状的场地滑冰。因为无人机足够的多（但是不会拥堵或者相撞），我们认为可以用一个概率密度函数 $\rho(t, v) (\geq 0)$ 来刻画无人机的分布。因为场地是圆环形的，所以我们可以认为速度 $v \in \mathbb{R}$ ，它表示无人机的线速度。那么，对于任意给定的时间 t ，和两个速度 $v_1 < v_2$ ，

$$\int_{v_1}^{v_2} \rho(t, v) dv$$

表示全体无人机中速度介于 v_1 和 v_2 之间的概率。

由于无人机的运动机理，已知这个密度函数的演化满足如下的方程

$$\rho_t + ((u(t) - v)\rho)_v = \rho_{vv}, \quad v \in \mathbb{R}, \quad t > 0.$$

其中 $u(t)$ 为无人机的指令速度。

(1) 为了研究怎么给无人机合适的指令速度，大宝给导演提议说，应该让

$$u(t) = u_0 + u_1 N(t)$$

其中 $u_0 > 0$ ， $u_1 > 0$ ，而 $N(t)$ 表示无人机的速度正部 ($v_+ = \max\{0, v\}$) 的平均

$$N(t) = \int_{-\infty}^{+\infty} v_+ \rho(t, v) dv = \int_0^{+\infty} v \rho(t, v) dv.$$

但是，你善意地提醒道，如果 $u_1 > 1$ 那么 $N(t)$ 在演化过程中不会有上界，以至于很快引起无人机的故障。你可以证明你的上述结论吗？（为了方便讨论，我们忽略 ρ 及其导数在 $|v| \rightarrow +\infty$ 时的贡献。）

(2) 在采纳了大宝和你的建议后，导演又在考虑这些无人机是否会在滑行中在圆圈场地上均匀分布，于是我们需要考虑关于速度和位置的联合密度函数 $p(t, x, v) (\geq 0)$ 。这里 $x \in [0, 2\pi]$ 表示无人机在圆圈上的相对位置，显然 $\int_0^{2\pi} p(t, x, v) dx = \rho(t, v)$ 。已知这个联合密度函数的演化满足

$$p_t + vp_x + ((u(t) - v)p)_v = p_{vv}, \quad x \in [0, 2\pi], \quad v \in \mathbb{R}, \quad t > 0.$$

且由于无人机在绕圈滑行，在 x 方向上如下条件满足

$$p(t, 0, v) = p(t, 2\pi, v), \quad v \in \mathbb{R}, \quad t > 0.$$

你大胆猜测：无论初始分布如何，无人机很快就在圆圈上接近一个均匀分布。你可以证明或者证伪这个命题吗？（为了方便讨论，我们忽略 p 及其导数在 $|v| \rightarrow +\infty$ 时的贡献。）

10 答案 (1) 我们定义平均速度

$$M(t) = \int_{-\infty}^{+\infty} v \rho(t, v) dv.$$

直接计算得

$$\begin{aligned} \frac{d}{dt} M(t) &= \frac{d}{dt} \int_{-\infty}^{+\infty} v \rho(t, v) dv \\ &= \int_{-\infty}^{+\infty} v \rho_t(t, v) dv \\ &= \int_{-\infty}^{+\infty} v (-(u(t) - v) \rho + \rho_v)_v dv \\ &= \int_{-\infty}^{+\infty} ((u(t) - v) \rho - \rho_v) dv \\ &= u(t) - M(t) \\ &= u_0 + u_1 N(t) - M(t). \end{aligned}$$

由于 $M(t) \leq N(t)$, 又显然 $N(t) \geq 0$, 我们得到

$$\frac{d}{dt} M(t) \geq u_0 + (u_1 - 1)N(t) \geq u_0 > 0.$$

则, 当 $t \rightarrow +\infty$, 我们有 $M(t) \rightarrow +\infty$ 。

而 $N(t) \geq M(t)$, 所以 $N(t)$ 也将发散到正无穷。

(2) 由于 x 方向的周期边界条件, 我们将方程的解写成如下傅里叶级数的形式

$$p(t, x, v) = \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} p_k(t, v) e^{ikx}.$$

其中

$$p_k(t, v) = \int_0^{2\pi} p(t, x, v) e^{-ikx} dx.$$

那么, 我们需要证明当 $k \neq 0$ 时, $p_k(t, v)$ 将衰减。

我们首先利用傅里叶级数的正交性, 可得 p_k 满足

$$\partial_t p_k + ikv p_k = -\partial_v ((u(t) - v) p_k)_v + \partial_{vv} p_k, \quad t > 0, \quad k \in \mathbb{Z}, \quad v \in \mathbb{R}.$$

为了进一步求解, 下面使用关于 v 的傅里叶变换。令

$$\hat{p}_k(t, \xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} p_k(t, v) e^{-iv\xi} dv$$

经过计算, 可以得到 $\hat{p}_k(t, \xi)$ 满足以下的方程

$$\partial_t \hat{p}_k + (\xi - k) \partial_\xi \hat{p}_k = - (i\xi u(t) + \xi^2) \hat{p}_k.$$

接下来, 我们使用特征线法求解此方程. 考虑如下特征线方程

$$\frac{d}{dt} \xi_k(t) = \xi_k(t) - k.$$

其通解为

$$\xi_k(t) - k = e^{t-s} (\xi_k(s) - k).$$

沿着特征线, 我们得到

$$\hat{p}_k(t, \xi_k(t)) = \hat{p}_k(0, \xi_k(0)) e^{-\int_0^t (\xi_k(s))^2 + iu(s) \xi_k(s) ds}$$

再利用特征线的通解, 我们可得

$$\hat{p}_k(t, \xi) = \hat{p}_k(0, (\xi - k)e^{-t} + k) e^{-H_{t,k}(\xi - k)},$$

其中

$$H_{t,k}(z) = \frac{1 - e^{-2t}}{2} z^2 + 2k(1 - e^{-t})z + i \int_0^t u(s) e^{-(t-s)} ds z + k^2 t + ik \int_0^t u(s) ds.$$

然后, 经过傅里叶逆变换, 我们可以得到 $p_k(t, v)$ 的表达式

$$p_k(t, v) = e^{ikv} (p_{t,k} * G_{t,k})(v).$$

其中, $p_{t,k}(y)$ 的定义如下

$$p_{t,k}(y) := e^t p_{0,k}(e^t y), \quad p_{0,k}(v) := e^{-ikv} \int_0^{2\pi} p_{\text{init}}(x, v) e^{-ikx} dx.$$

而 $G_{t,k}(z)$ 是一个复值的高斯函数, 表达式如下

$$G_{t,k}(z) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(z - \mu)^2}{2\sigma^2}\right) \exp(ik\Theta) \exp(-k^2 D).$$

这里,

$$\begin{aligned} \mu(t) &= \int_0^t e^{-(t-s)} u(s) ds, \quad \sigma(t) = \sqrt{1 - e^{-2t}}, \\ \Theta(t, z) &= -\frac{2(z - \mu(t))}{1 - e^{-t}} - \int_0^t u(s) ds, \quad D(t) = t - 2\frac{1 - e^{-t}}{1 + e^{-t}}. \end{aligned}$$

我们注意到, 当 $t > 0$ 时, 高斯函数 $G_{t,k}$ 由于衰减因子 $\exp(-k^2 D)$ 的存在将衰减, 这个衰减速率对于 t 充分大时是指数型的. 最后, 利用卷积不等式, 我们可以得到

$$\begin{aligned} \|p_k(t, v)\|_{L^1(\mathbb{R})} &= \|e^{ikv} (p_{t,k} * G_{t,k})(v)\|_{L^1(\mathbb{R})} \leq \|p_{t,k}(g)\|_{L^1(\mathbb{R})} \|G_{t,k}\|_{L^1(\mathbb{R})} \\ &= \|p_{0,k}(g)\|_{L^1(\mathbb{R})} e^{-k^2 D(t)}. \end{aligned}$$

这证明了, 当 $k \neq 0$ 时, $p_k(t, v)$ 将迅速衰减, 因而 $p(t, x, v)$ 将充分接近 $\frac{1}{\pi} p_0(t, v)$, 这对应了一个空间方向的均匀分布.

2022 Alibaba Global Mathematics Competition

Single-Choice Problem: Magic Magnetic Cube

Divide a solid cube $ABCD - A_1B_1C_1D_1$ (with $AB = 1$) into 12 pieces (Figure 1) as follows:

- 1) Take 6 diagonals of its surfaces $AC, AB_1, AD_1, C_1B, C_1D, C_1A_1$;
- 2) Consider all triangles with the center of the cube as a vertex, and one of the above 6 diagonals and 12 edges as the opposite side;
- 3) These 18 triangles cut the cube into 12 tetrahedra, and each tetrahedron has two edges that are *cube edges*;
- 4) Each tetrahedron is connected to other tetrahedra **only** by its two cube edges.

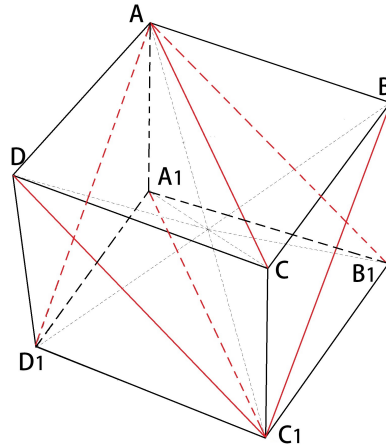


Figure 1: Magic magnetic cube

Such a toy can take on a variety of shapes (Figure 2).

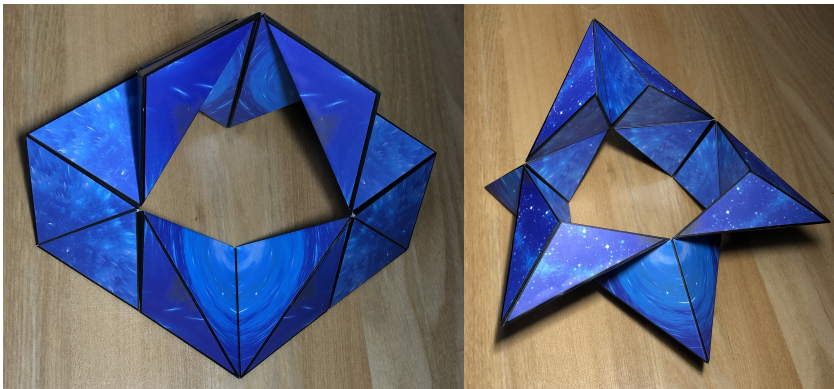


Figure 2: Examples of various shapes

Question: Of all the possible shapes of this toy, what is the maximum distance (in space) between two points on it?

$\sqrt{11}$ $\sqrt{7+4\sqrt{2}}$ $\sqrt{13}$ $1+2\sqrt{2}$ None of the above

Single-Choice Problem: Onlook With Distance

One day, there is a Street Art Show at somewhere, and there are some spectators around. We consider this place as an Euclidean plane. Let K be the center of the show. And name the spectators by $A_1, A_2, \dots, A_n, \dots$. They pick their positions

$$P_1, P_2, \dots, P_n, \dots,$$

one by one. The positions need to satisfy the following three conditions simultaneously.

- (i) The distance between K and A_n is no less than 10 meters, that is, $KP_n \geq 10\text{m}$ holds for any positive integer n .
- (ii) The distance between A_n and any previous spectator is no less than 1 meter, that is, $P_m P_n \geq 1\text{m}$ holds for any $n \geq 2$ and any $1 \leq m \leq n - 1$.
- (iii) A_n always choose the position closest to K that satisfies (i) and (ii), that is, KP_n reaches its minimum possible value. If there are more than one point that satisfy (i) and (ii) and have the minimum distance to K , A_n may choose any one of them.

For example, A_1 is not restricted by (ii), so he may choose any point on the circle C which is centered at K with radius 10 meters. For A_2 , since there are lots of points on C which are at least 1 meter apart from P_1 , he may choose anyone of them.

(1) Which of the following statement is true?

There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1 \leq KP_n \leq c_2$ always hold (unit: meter);

There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1\sqrt{n} \leq KP_n \leq c_2\sqrt{n}$ always hold (unit: meter);

There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1n \leq KP_n \leq c_2n$ always hold (unit: meter);

There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1n^2 \leq KP_n \leq c_2n^2$ always hold (unit: meter).

(2) Since human bodies are 3-dimensional, if one spectator's position is near another spectator's path of view, then the second one's sight will be blocked by the first one. Suppose that for different i, j , if the circle centered at P_i with radius $\frac{1}{6}$ meter intersects with segment KP_j , then A_j 's sight will be blocked by A_i , and A_j could not see the entire show.

Which of the following statement is true?

If there were 60 spectators, then some of them could not see the entire show;

If there were 60 spectators, then it is possible that all spectators could see the entire show, but if there were 800 spectators, then some of them could not see the entire show;

If there were 800 spectators, then it is possible that all spectators could see the entire show, but if there were 10000 spectators, then some of them could not see the entire show;

If there were 10000 spectators, then it is possible that all spectators could see the entire show.

Single-Choice Problem: Tiger Mystery Box

Brave NiuNiu (a milk drink company) organizes a promotion during the Chinese New Year: one gets a red packet when buying a carton of milk of their brand, and there is one of the following characters in the red packet “虎”(Tiger), “生”(Gain), “威”(Strength).



If one collects two “虎”, one “生” and one “威”, then they form a Chinese phrases “虎虎生威”(Pronunciation: hu hu sheng wei), which means “Have the courage and strength of the tiger”. This is a nice blessing because the Chinese zodiac sign for the year 2022 is tiger. Soon, the product of Brave NiuNiu becomes quite popular and people hope to get a collection of “虎虎生威”. Suppose that the characters in every packet are independently random, and each character has probability $\frac{1}{3}$.

(1) What is the expectation of cartons of milk to collect “虎虎生威”(i.e. one collects at least 2 copies of “虎”, 1 copy of “生”, 1 copy of “威”)?

$6\frac{1}{3}$ $7\frac{1}{3}$ $8\frac{1}{3}$ $9\frac{1}{3}$ None of the above

(2) In a weekly meeting of Brave NiuNiu, its market team notices that one often has to collect too many “生” and “威”, before getting a collection of “虎虎生威”. Thus an improved plan is needed for the proportion of characters. Suppose that the probability distribution of “虎”, “生” and “威” is (p, q, r) , then which of the following plans has the smallest expectation (among the 4) for a collection of “虎虎生威”?

$$(p, q, r) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$$

$$(p, q, r) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$$

$$(p, q, r) = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$$

$$(p, q, r) = (\frac{3}{4}, \frac{1}{8}, \frac{1}{8})$$

Given a set X and a function $f : X \times X \rightarrow [0, 1]$, we say f is right uniform if for any $\epsilon > 0$, there exist finitely many elements $b_1, b_2, \dots, b_m$ in X such that for any $t \in X$, the following holds for some $b_{i(t)}$:

$$|f(x, t) - f(x, b_{i(t)})| < \epsilon, \forall x \in X.$$

Similarly, we say f is left uniform if for any $\epsilon > 0$, there exist finitely many elements $a_1, a_2, \dots, a_n$ in X such that for any $t \in X$, the following holds for some $a_{i(t)}$:

$$|f(t, x) - f(a_{i(t)}, x)| < \epsilon, \forall x \in X.$$

Prove that f is right uniform if and only if it's left uniform.

Let n be a positive integer and $V = \mathbb{R}^n$ be an n -dimensional Euclidean space with a basis $e_i = (\underbrace{0, \dots, 0}_{i-1}, 1, \underbrace{0, \dots, 0}_{n-i})$ ($1 \leq i \leq n$) and with an inner product $(\cdot, \cdot)$ defined by

$$(e_i, e_j) = \delta_{i,j}$$

where

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is Kronecker's symbol. For a nonzero vector $v \in V$, define $s_v : V \rightarrow V$ by

$$s_v(u) = u - \frac{2(u, v)}{(v, v)}v, \quad \forall u \in V.$$

For an integer k between 0 and n , write $\text{Gr}_k(V)$ for the set of k -dimensional subspaces of V . For a k -dimensional subspace W of V , write $[W]$ for the corresponding element of $\text{Gr}_k(V)$. Choose an orthonormal basis $\{v_1, \dots, v_k\}$ of W , define $s_{[W]} : V \rightarrow V$ by

$$s_{[W]} = s_{v_1} \cdots s_{v_k}.$$

- (1) Prove that $s_{[W]}$ is independent of the choice of an orthonormal basis $\{v_1, \dots, v_k\}$.
- (2) Prove that $s_{[W]}^2 = \text{id}$.
- (3) For another element $[W'] \in \text{Gr}_k(V)$, define

$$t_{[W]}([W']) = [s_{[W]}(W')],$$

where $s_{[W]}(W')$ is the image of W' under $s_{[W]}$. We call a subset X of $\text{Gr}_k(V)$ a “nice set” if

$$t_{[W]}([W']) = [W'], \quad \forall [W], [W'] \in X.$$

Find the maximal cardinality of a “nice set” in $\text{Gr}_k(V)$ and prove it.

A Busy Courier

A courier picks up a package at coordinate $(n, 0)$ in the two-dimensional lattice, while his station locates at the origin $(0, 0)$. The courier then does a discrete time simple random walk on $\mathbb{Z}^2$. In the rest of this question, you may without loss of generality assume n is sufficiently large.

- (1) Let $P_{1,n}$ be the probability that at his $\lfloor n^{1.5} \rfloor$ th step, the distance between this courier and his station is greater than $\frac{n}{2}$. Prove that

$$\lim_{n \rightarrow +\infty} P_{1,n} = 1$$

- (2) Let $P_{2,n}$ be the probability that the courier has ever reached the station within his first $\lfloor n^{1.5} \rfloor$ steps. Prove that

$$\lim_{n \rightarrow +\infty} P_{2,n} = 0$$

- (3) Let $P_{3,n}$ be the probability that the courier has ever reached the station within his first 2^n steps. Prove that

$$\lim_{n \rightarrow +\infty} P_{3,n} = 1$$

- (1) Show that there is no *periodic* sequence $a_1, a_2, a_3, \dots$ of signs $a_n \in \{\pm 1\}$ such that, for every *rational* number θ ,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

- (2) Show that there is no sequence $a_1, a_2, a_3, \dots$ of signs $a_n \in \{\pm 1\}$ such that, for every *rational* number θ

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq 2022?$$

- (3) Give an example of a sequence of signs $a_n \in \{\pm 1\}$ such that, for every $\theta \in \mathbb{Q} - \mathbb{Z}$,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty,$$

and which takes each of the values $+1$ and -1 an infinite number of times.

Program Design For The Opening Ceremony

Suppose you are chosen as a technology assistant by the director of the opening ceremony for the 2022 Winter Olympics, and your job is to evaluate the program proposals. One of the backup programs is a skating show of a ensemble of drones dressed as mascots, which are moving along a circle. Since the number of the drones is sufficiently large, we can use a probability density function $\rho(t, v) (\geq 0)$ to represent the distribution of the drones. Here, $v \in \mathbb{R}$ is the line speed, and for a given time t , and two speeds $v_1 < v_2$,

$$\int_{v_1}^{v_2} \rho(t, v) dv$$

is the probability of find a drone with its speed between v_1 and v_2 .

Suppose that the dynamics of the density function is governed by

$$\rho_t + ((u(t) - v)\rho)_v = \rho_{vv}, \quad v \in \mathbb{R}, \quad t > 0,$$

where $u(t)$ is the command speed.

(1) To investigate proper choices of the command speed, D.B. propose that we shall choose

$$u(t) = u_0 + u_1 N(t)$$

where $u_0 > 0$, $u_1 > 0$ and $N(t)$ is the average of the positive part of the speed $v_+ = \max\{0, v\}$, i.e.,

$$N(t) = \int_{-\infty}^{+\infty} v_+ \rho(t, v) dv = \int_0^{+\infty} v \rho(t, v) dv.$$

But you claim that if $u_1 > 1$, $N(t)$ may become unbounded in evolution, such that the drones may be out of order. Can you prove it? (For simplicity, the contributions of ρ and its derivatives at $|v| \rightarrow +\infty$ are neglected.)

(2) After taking those advices, the directly is wondering whether the drones will be evenly distributed along the circle. Thus, we shall consider the joint density function $p(t, x, v) (\geq 0)$ of position and speed, where $x \in [0, 2\pi]$ is the position coordinate on the circle. Clearly, $\int_0^{2\pi} p(t, x, v) dx = \rho(t, v)$. Suppose that the governing equation for $p(t, x, v)$ is

$$p_t + vp_x + ((u(t) - v)p)_v = p_{vv}, \quad x \in [0, 2\pi], \quad v \in \mathbb{R}, \quad t > 0.$$

Because, the drones are circulating around, the following boundary condition is satisfied

$$p(t, 0, v) = p(t, 2\pi, v), \quad v \in \mathbb{R}, \quad t > 0.$$

You have a feeling that, no matter how the drones are distributed initially, they will become almost evenly distributed very quickly. Can you prove or disprove this statement? (For simplicity, the contributions of p and its derivatives at $|v| \rightarrow +\infty$ are neglected.)

2022 Alibaba Global Mathematics Competition (Qualifying Round)

1 (Single-Choice Problem) Magic Magnetic Cube

Divide a solid cube $ABCD - A_1B_1C_1D_1$ (with $AB = 1$) into 12 pieces (Figure 1) as follows:

- 1) Take 6 diagonals of its surfaces $AC, AB_1, AD_1, C_1B, C_1D, C_1A_1$;
- 2) Consider all triangles with the center of the cube as a vertex, and one of the above 6 diagonals and 12 edges as the opposite side;
- 3) These 18 triangles cut the cube into 12 tetrahedra, and each tetrahedron has two edges that are *cube edges*;
- 4) Each tetrahedron is connected to other tetrahedra **only** by its two cube edges.

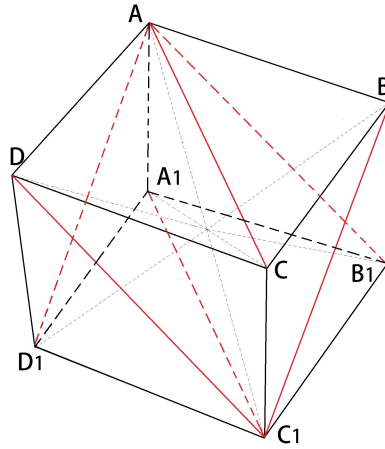


Figure 1: Magic magnetic cube

Such a toy can take on a variety of shapes (Figure 2).

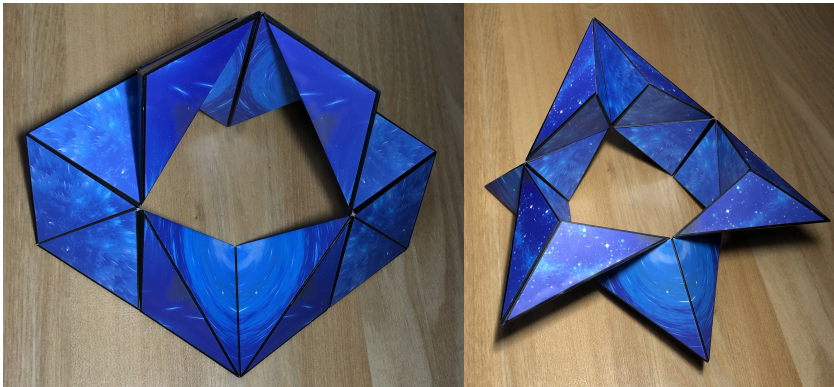


Figure 2: Examples of various shapes

Question: Of all the possible shapes of this toy, what is the maximum distance (in space) between two points on it?

- (A) $\sqrt{11}$ (B) $\sqrt{7 + 4\sqrt{2}}$ (C) $\sqrt{13}$ (D) $1 + 2\sqrt{2}$ (E) None of the above

1 Answer The answer is **D**. Label the eight vertices of the cube with $ABCD-A_1B_1C_1D_1$, and let the pair of vertices selected at the beginning be A and C_1 . Then, these two vertices belong to 6 small tetrahedrons, and B, C, D, D_1, A_1, B_1 (no matter how they are deformed) constitute a Hexagon with side length 1 (possibly degenerate).

Note that this toy consists of 12 tetrahedra (and thus convex), so the maximum value must be the vertices of the two small tetrahedra.

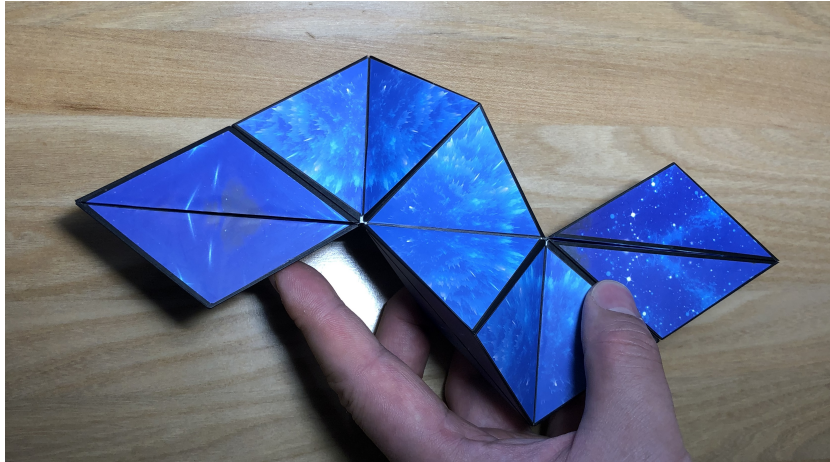
In a tetrahedron, the four vertices can be divided into 3 categories:

- A vertex corresponds to A or C_1 in the cube, we call it a vertex of **Type I**;
- The two vertices (and thus the edges between them) belong to the (spatial) hexagon $BCDD_1A_1B_1$, which we call vertices of **Type II**;
- A vertex corresponds to the center of the cube, and its distance to the other three vertices of the same small tetrahedron is $\frac{\sqrt{3}}{2}$, we call it a vertex of **Type III**.

We examine the maximum value of the distance **along the edges** between these vertices, and it is easy to get the following table:

distance	vertices of Type I	vertices of Type II	vertices of Type III
first kind	$1 + 2\sqrt{2}$	$2 + \sqrt{2}$	$1 + \sqrt{2} + \frac{\sqrt{3}}{2}$
second kind	$2 + \sqrt{2}$	3	$2 + \frac{\sqrt{3}}{2}$
third kind	$1 + \sqrt{2} + \frac{\sqrt{3}}{2}$	$2 + \frac{\sqrt{3}}{2}$	$2 + \sqrt{3}$

Therefore, the distance between any two points on this toy will not exceed $1 + 2\sqrt{2}$. And this value can be achieved, as shown in the following figure:



2,3 (Single-Choice Problem) Onlook With Distance

One day, there is a Street Art Show at somewhere, and there are some spectators around. We consider this place as an Euclidean plane. Let K be the center of the show. And name the spectators by $A_1, A_2, \dots, A_n, \dots$. They pick their positions

$$P_1, P_2, \dots, P_n, \dots,$$

one by one. The positions need to satisfy the following three conditions simultaneously.

- (i) The distance between K and A_n is no less than 10 meters, that is, $KP_n \geq 10\text{m}$ holds for any positive integer n .
- (ii) The distance between A_n and any previous spectator is no less than 1 meter, that is, $P_m P_n \geq 1\text{m}$ holds for any $n \geq 2$ and any $1 \leq m \leq n - 1$.
- (iii) A_n always choose the position closest to K that satisfies (i) and (ii), that is, KP_n reaches its minimum possible value. If there are more than one point that satisfy (i) and (ii) and have the minimum distance to K , A_n may choose any one of them.

For example, A_1 is not restricted by (ii), so he may choose any point on the circle C which is centered at K with radius 10 meters. For A_2 , since there are lots of points on C which are at least 1 meter apart from P_1 , he may choose anyone of them.

(1) Which of the following statement is true?

- (A) There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1 \leq KP_n \leq c_2$ always hold (unit: meter);
- (B) There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1\sqrt{n} \leq KP_n \leq c_2\sqrt{n}$ always hold (unit: meter);
- (C) There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1n \leq KP_n \leq c_2n$ always hold (unit: meter);
- (D) There exist positive real numbers c_1, c_2 such that for any positive integer n , no matter how $A_1, A_2, \dots, A_n$ choose their positions, $c_1n^2 \leq KP_n \leq c_2n^2$ always hold (unit: meter).

(2) Since human bodies are 3-dimensional, if one spectator's position is near another spectator's path of view, then the second one's sight will be blocked by the first one. Suppose that for different i, j , if the circle centered at P_i with radius $\frac{1}{6}$ meter intersects with segment KP_j , then A_j 's sight will be blocked by A_i , and A_j could not see the entire show.

Which of the following statement is true?

- (A) If there were 60 spectators, then some of them could not see the entire show;
- (B) If there were 60 spectators, then it is possible that all spectators could see the entire show, but if there were 800 spectators, then some of them could not see the entire show;

- (C) If there were 800 spectators, then it is possible that all spectators could see the entire show, but if there were 10000 spectators, then some of them could not see the entire show;
- (D) If there were 10000 spectators, then it is possible that all spectators could see the entire show.

2 Answer The answer is **B**. Suppose the length of KP_n is d_n meters.

We consider the discs centered at $P_1, P_2, \dots, P_{n-1}$ with radius 1 meter. Use the property of P_n we get that these discs and the interior of C cover the disc centered at K with radius d_n , so

$$\pi \cdot d_n^2 \leq (n-1) \cdot \pi \cdot 1^2 + \pi \cdot 10^2,$$

It follows that

$$d_n \leq \sqrt{n+99} \leq \sqrt{100n} = 10\sqrt{n}.$$

On the other hand, we consider the discs centered at $P_1, P_2, \dots, P_n$ with radius $\frac{1}{2}$ meter. Since the distance between any two of $P_1, P_2, \dots, P_n$ is not less than 1 meter, all these discs do not intersect.

Note that every length of $KP_1, KP_2, \dots, KP_n$ is not more than d_n meter (If the length of KP_m is more than d_n meter, then A_m can choose P_n , which is closer to K , contradiction.)

So all these discs are inside the circle centered at K with radius $d_n + \frac{1}{2}$, and

$$\pi(d_n + \frac{1}{2})^2 \geq n \cdot \pi \cdot (\frac{1}{2})^2,$$

So

$$d_n \geq \frac{\sqrt{n}}{2} - \frac{1}{2}.$$

For $n = 1$, $d_1 = 10$; For $n \geq 2$, we have that $\frac{1}{2} < \frac{2\sqrt{n}}{5}$, so

$$d_n > \frac{\sqrt{n}}{2} - \frac{2\sqrt{n}}{5} = \frac{\sqrt{n}}{10}.$$

Therefore, $\frac{\sqrt{n}}{10} \leq d_n \leq 10\sqrt{n}$, (B) is correct.

3 Answer The answer is **B**. For 60 residents, since $\sin \frac{\pi}{60} > \frac{1}{10} \sin \frac{\pi}{6} = \frac{1}{20}$, the side length of a regular 60-gon inscribed in C is not less than 1 meter. Therefore, $P_1, P_2, \dots, P_{60}$ may be all vertices of this polygon. For different i, j , the distance from P_i to KP_j is not less than $10 \sin \frac{\pi}{30}$ meter. Since $\sin \frac{\pi}{30} > \frac{1}{5} \sin \frac{\pi}{6} = \frac{1}{10}$, all residents could see the entire show.

On the other hand, if there are 800 residents, we draw rays $\overrightarrow{KP_1}, \overrightarrow{KP_2}, \dots, \overrightarrow{KP_{800}}$. If two of them (called $\overrightarrow{KP_i}, \overrightarrow{KP_j}$) coincide, and $KP_j > KP_i$, then A_j 's sight line is blocked by A_i .

Suppose no two rays coincide, we first prove that if an angle $\angle P_i K P_j$ satisfies $\angle P_i K P_j \leq \frac{1}{12} \left(\frac{1}{K P_i} + \frac{1}{K P_j} \right)$ (the unit of angle is rad, and the unit of length is meter), then the sight line of one of A_i, A_j is blocked by the other.

Without loss of generality we suppose $K P_i \leq K P_j$. Since $K P_i, K P_j \geq 10$, we get that $\angle P_i K P_j \leq \frac{1}{60}$, so it is acute. Therefore, the foot point from P_i to $K P_j$ is inside segment $K P_j$, and the distance from P_i to $K P_j$ is

$$K P_i \sin \angle P_i K P_j < K P_i \cdot \angle P_i K P_j \leq \frac{K P_i}{12} \left(\frac{1}{K P_i} + \frac{1}{K P_j} \right) \leq \frac{1}{6},$$

So, A_j 's sight line is blocked by A_i .

Note that $\overrightarrow{K P_1}, \overrightarrow{K P_2}, \dots, \overrightarrow{K P_{800}}$ cut the perigon with vertex K to 800 angles, and their sum is 2π , but the sum of all $\frac{1}{12} \left(\frac{1}{K P_i} + \frac{1}{K P_j} \right)$ is

$$\begin{aligned} \sum \frac{1}{12} \left(\frac{1}{K P_i} + \frac{1}{K P_j} \right) &= \frac{1}{6} \sum_{i=1}^{800} \frac{1}{K P_i} \\ &\geq \frac{1}{6} \sum_{i=1}^{800} \frac{1}{\sqrt{i+99}} \quad (\text{from the conclusion of (1)}) \\ &= \frac{1}{6} \sum_{m=100}^{899} \frac{1}{\sqrt{m}} \\ &\geq \frac{1}{6} \sum_{m=100}^{899} \left(\int_m^{m+1} \frac{1}{\sqrt{x}} dx \right) \quad (\text{since } \frac{1}{\sqrt{x}} \text{ is a decreasing function}) \\ &= \frac{1}{6} \int_{100}^{900} \frac{1}{\sqrt{x}} dx = \frac{\sqrt{900} - \sqrt{100}}{3} = \frac{20}{3} > 2\pi. \end{aligned}$$

Therefore, there exist an angle $\angle P_i K P_j$ satisfies $\angle P_i K P_j \leq \frac{1}{12} \left(\frac{1}{K P_i} + \frac{1}{K P_j} \right)$, and some of the residents could not see the entire show.

4,5 (Single-Choice Problem) Tiger Mystery Box

Brave NiuNiu (a milk drink company) organizes a promotion during the Chinese New Year: one gets a red packet when buying a carton of milk of their brand, and there is one of the following characters in the red packet “虎”(Tiger), “生”(Gain), “威”(Strength).



If one collects two “虎”, one “生” and one “威”, then they form a Chinese phrases “虎虎生威”(Pronunciation: hu hu sheng wei), which means “Have the courage and strength of the tiger”. This is a nice blessing because the Chinese zodiac sign for the year 2022 is tiger. Soon, the product of Brave NiuNiu becomes quite popular and people hope to get a collection of “虎虎生威”. Suppose that the characters in every packet are independently random, and each character has probability $\frac{1}{3}$.

(1) What is the expectation of cartons of milk to collect “虎虎生威”(i.e. one collects at least 2 copies of “虎”, 1 copy of “生”, 1 copy of “威”)?

- (A) $6\frac{1}{3}$
- (B) $7\frac{1}{3}$
- (C) $8\frac{1}{3}$
- (D) $9\frac{1}{3}$
- (E) None of the above

(2) In a weekly meeting of Brave NiuNiu, its market team notices that one often has to collect too many “生” and “威”, before getting a collection of “虎虎生威”. Thus an improved plan is needed for the proportion of characters. Suppose that the probability distribution of “虎”, “生” and “威” is (p, q, r) , then which of the following plans has the smallest expectation (among the 4) for a collection of “虎虎生威”?

- (A) $(p, q, r) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
- (B) $(p, q, r) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$
- (C) $(p, q, r) = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$
- (D) $(p, q, r) = (\frac{3}{4}, \frac{1}{8}, \frac{1}{8})$

4 Answer The answer is **B**. We can use Poisson process to get the explicit formula for the general case. Suppose that there are in total n characters. The probability for the character i is p_i , and we aim to collect k_i copies of the character i . We denote by N the first time to realize our collection, and we need to calculate $\mathbb{E}[N]$. A nice technique is embedding this model to a Poisson process: we have a Poisson process of density 1. Every time when the signal arrives, we sample independently p_i for the character i . We also denote by

$$T_i = \inf\{t \in \mathbb{R}_+ : \text{before } t \text{ one collects } k_i \text{ copies of the character } i\},$$

$$T = \max_{1 \leq i \leq n} T_i.$$

We claim that $\mathbb{E}[T] = \mathbb{E}[N]$. Let us prove this claim. We denote by τ_j the waiting time for the j -th signal, then we have

$$T = \sum_{j=1}^N \tau_j.$$

By the property of Poisson process, $(\tau_j)_{j \geq 1}$ are i.i.d. exponential random variable. Using conditional probability we have

$$\begin{aligned} \mathbb{E}[T] &= \mathbb{E}\left[\sum_{j=1}^N \tau_j\right] \\ &= \sum_{k=0}^{\infty} \mathbb{E}\left[\sum_{j=1}^k \tau_j \mid N = k\right] \mathbb{P}[N = k] \\ &= \sum_{k=0}^{\infty} \mathbb{P}[N = k] k \mathbb{E}[\tau_1] \\ &= \mathbb{E}[N]. \end{aligned}$$

This justifies our claim.

Then it suffices to calculate $\mathbb{E}[T]$. By Fubini's lemma

$$\begin{aligned} \mathbb{E}[T] &= \int_0^{\infty} \mathbb{P}[T > t] dt \\ &= \int_0^{\infty} (1 - \mathbb{P}[T \leq t]) dt \\ &= \int_0^{\infty} (1 - \mathbb{P}[T_i \leq t, \forall 1 \leq i \leq n]) dt. \end{aligned}$$

In fact, the thinning property of the Poisson process creates a lot of independence. We can treat the collection of the character i as independent Poisson processes of parameter p_i . Then we have

$$\mathbb{E}[T] = \int_0^{\infty} \left(1 - \prod_{i=1}^n \mathbb{P}[T_i \leq t]\right) dt.$$

We write down directly the explicit formula of $\mathbb{P}[T_i \leq t]$ using Poisson distribution

$$\mathbb{P}[T_i \leq t] = 1 - \sum_{k=0}^{k_i-1} e^{-p_i t} \frac{(p_i t)^k}{k!}.$$

Finally, it gives us

$$\mathbb{E}[N] = \int_0^\infty \left(1 - \prod_{i=1}^n \left(1 - \sum_{k=0}^{k_i-1} e^{-p_i t} \frac{(p_i t)^k}{k!} \right) \right) dt.$$

In our setting, $n = 3$, and the object $(k_1, k_2, k_3) = (2, 1, 1)$. Thus we have

$$\mathbb{E}[N] = 1 + p_1 + \left(\frac{2}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} \right) - \sum_{i=1}^3 \frac{1}{1-p_i} - \frac{p_1}{(p_1+p_2)^2} - \frac{p_1}{(p_1+p_3)^2}.$$

When $(p_1, p_2, p_3) = (1/3, 1/3, 1/3)$, the expectation is $7\frac{1}{3}$.

5 Answer The answer is **C**. In last question, we know the expectation for Plan A is $7\frac{1}{3}$. Plan D is not a good plan obviously, because the expectation to collect “威” is 8, which is larger than Plan A. It suffices to calculate Plan B and C. Using the expression

$$\begin{aligned} \mathbb{E}[\tau] = 1 + p + \left(\frac{2}{p} + \frac{1}{q} + \frac{1}{r} \right) - \left(\frac{1}{p+q} + \frac{1}{p+r} + \frac{1}{q+r} \right) \\ - \frac{p}{(p+q)^2} - \frac{p}{(p+r)^2}. \end{aligned}$$

The expectation for Plan B and Plan C are respectively $7\frac{1}{18}, 6\frac{223}{245}$. Plan C is the best one.

6 (Proof Question)

Given a set X and a function $f : X \times X \rightarrow [0, 1]$, we say f is right uniform if for any $\epsilon > 0$, there exist finitely many elements $b_1, b_2, \dots, b_m$ in X such that for any $t \in X$, the following holds for some $b_{i(t)}$:

$$|f(x, t) - f(x, b_{i(t)})| < \epsilon, \forall x \in X.$$

Similarly, we say f is left uniform if for any $\epsilon > 0$, there exist finitely many elements $a_1, a_2, \dots, a_n$ in X such that for any $t \in X$, the following holds for some $a_{i(t)}$:

$$|f(t, x) - f(a_{i(t)}, x)| < \epsilon, \forall x \in X.$$

Prove that f is right uniform if and only if it's left uniform.

6 Answer It suffices to prove that if f is right uniform, then it's left uniform. Given $\epsilon > 0$, we have finitely many elements $b_1, b_2, \dots, b_m$ as in the definition of being right uniform. Consider the map $h : X \rightarrow [0, 1]^m$, $h(x) = (f(x, b_1), f(x, b_2), \dots, f(x, b_m))$. Because $[0, 1]^m$ is compact, there are $c_1, c_2, \dots, c_n$ in X such that for any $x \in X$, $|h(x) - h(c_{i_x})| < \epsilon$ holds for some c_{i_x} . Hence

$$|f(x, b_i) - f(c_{i_x}, b_i)| < \epsilon, \forall i \in \{1, 2, \dots, m\}. \quad (1)$$

Now since f is right uniform, we have $|f(x, t) - f(x, b_{i_t})| < \epsilon (\forall x \in X)$, $|f(c_{i_x}, t) - f(c_{i_x}, b_{i_t})| < \epsilon$. Combined with (1), it follows that for any $t \in X$,

$$|f(x, t) - f(c_{i_x}, t)| \leq |f(x, t) - f(x, b_{i_t})| + |f(x, b_{i_t}) - f(c_{i_x}, b_{i_t})| + |f(c_{i_x}, b_{i_t}) - f(c_{i_x}, t)| < 3\epsilon.$$

Since ϵ is arbitrary, f is left uniform.

7 (Proof Questions)

Let n be a positive integer and $V = \mathbb{R}^n$ be an n -dimensional Euclidean space with a basis $e_i = (\underbrace{0, \dots, 0}_{i-1}, 1, \underbrace{0, \dots, 0}_{n-i})$ ($1 \leq i \leq n$) and with an inner product $(\cdot, \cdot)$ defined by

$$(e_i, e_j) = \delta_{i,j}$$

where

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is Kronecker's symbol. For a nonzero vector $v \in V$, define $s_v : V \rightarrow V$ by

$$s_v(u) = u - \frac{2(u, v)}{(v, v)}v, \quad \forall u \in V.$$

For an integer k between 0 and n , write $\text{Gr}_k(V)$ for the set of k -dimensional subspaces of V . For a k -dimensional subspace W of V , write $[W]$ for the corresponding element of $\text{Gr}_k(V)$. Choose an orthonormal basis $\{v_1, \dots, v_k\}$ of W , define $s_{[W]} : V \rightarrow V$ by

$$s_{[W]} = s_{v_1} \cdots s_{v_k}.$$

- (1) Prove that $s_{[W]}$ is independent of the choice of an orthonormal basis $\{v_1, \dots, v_k\}$.
- (2) Prove that $s_{[W]}^2 = \text{id}$.
- (3) For another element $[W'] \in \text{Gr}_k(V)$, define

$$t_{[W]}([W']) = [s_{[W]}(W')],$$

where $s_{[W]}(W')$ is the image of W' under $s_{[W]}$. We call a subset X of $\text{Gr}_k(V)$ a “nice set” if

$$t_{[W]}([W']) = [W'], \quad \forall [W], [W'] \in X.$$

Find the maximal cardinality of a “nice set” in $\text{Gr}_k(V)$ and prove it.

7 Answer (1) Write $W^\perp$ for the orthogonal complement of W in V . Then,

$$s_{[W]}|_W = 1 \text{ and } s_{[W]}|_{W^\perp} = -1.$$

This characterizes $s_{[W]}$ and shows its independence with the choice of an orthonormal basis of W .

(2) Since

$$s_{[W]}|_W = 1 \text{ and } s_{[W]}|_{W^\perp} = -1,$$

then $s_{[W]}^2 = \text{id}$.

(3) (i) We show that

$$t_{[W]}([W']) = [W'] \Leftrightarrow W' = (W' \cap W) \oplus (W' \cap W^\perp).$$

The sufficiency is clear. For the necessity, suppose that $t_{[W]}([W']) = [W']$. Then, for any $u \in W'$, we have $s_{[W]}(u) \in W'$. Write $u = u_1 + u_2$ where $u_1 \in W$ and $u_2 \in W^\perp$. Then, $u_1 - u_2 = s_{[W]}(u) \in W'$. Thus, $u_1 \in W'$ and $u_2 \in W'$. Hence,

$$W' \subset (W' \cap W) \oplus (W' \cap W^\perp).$$

It is clear that $(W' \cap W) \oplus (W' \cap W^\perp) \subset W'$. Therefore,

$$W' = (W' \cap W) \oplus (W' \cap W^\perp).$$

(ii) Put

$$X = \{[\text{span}_{\mathbb{R}}\{e_{i_1}, \dots, e_{i_k}\}] : 1 \leq i_1 < \dots < i_k \leq n\}.$$

By the above criterion in (i), X is a nice set in $\text{Gr}_k(V)$. This nice set X has cardinality $\binom{k}{n}$.

(iii) We prove by induction on n that any nice set X in $\text{Gr}_k(V)$ has cardinality at most $\binom{k}{n}$. When $n = 1$, this is trivial. Suppose the conclusion holds when $n < m$. Assume that $n = m$. When $k = 0$ or m , the conclusion is trivial. Assume that $1 \leq k \leq m - 1$. Choose an element $[W] \in X$. For each i ($0 \leq i \leq k$), write

$$X_i = \{[W'] \in X : \dim W' \cap W = k - i\}.$$

For each $[W'] \in X_i$, put

$$Y_i = \{[W' \cap W] : [W'] \in X_i\} \subset \text{Gr}_{k-i}(W)$$

and

$$Z_i = \{[W' \cap W^\perp] : [W'] \in X_i\} \subset \text{Gr}_i(W^\perp).$$

By the criterion in (i), Y_i is a nice set in $\text{Gr}_{k-i}(W)$ and Z_i is a nice set in $\text{Gr}_i(W^\perp)$. Since $\dim W = k < m$ and $\dim W^\perp = m - k < m$, by induction we get

$$|Y_i| \leq \binom{k-i}{k} \text{ and } |Z_i| \leq \binom{i}{m-k}.$$

Therefore,

$$|X| = \sum_{0 \leq i \leq k} |X_i| \leq \sum_{0 \leq i \leq k} |Y_i| |Z_i| \leq \sum_{0 \leq i \leq k} \binom{k-i}{k} \binom{i}{m-k} = \binom{k}{m}.$$

This shows the conclusion.

8 A Busy Courier

A courier picks up a package at coordinate $(n, 0)$ in the two-dimensional lattice and then does a discrete time simple random walk on $\mathbb{Z}^2$. His station locates at the origin $(0, 0)$. In the rest of this question, you may without loss of generality assume n is sufficiently large.

- (1) Let $P_{1,n}$ be the probability that at his $\lfloor n^{1.5} \rfloor$ th step, the distance between this courier and his station is greater than $\frac{n}{2}$. Prove that

$$\lim_{n \rightarrow +\infty} P_{1,n} = 1$$

- (2) Let $P_{2,n}$ be the probability that the courier has ever reached the station within his first $\lfloor n^{1.5} \rfloor$ steps. Prove that

$$\lim_{n \rightarrow +\infty} P_{2,n} = 0$$

- (3) Let $P_{3,n}$ be the probability that the courier has ever reached the station within his first 2^n steps. Prove that

$$\lim_{n \rightarrow +\infty} P_{3,n} = 1$$

8 Answer (1) Consider the event that this courier is of a distance no more than $\frac{n}{2}$ from the station after $\lfloor n^{1.5} \rfloor$ steps. By triangular inequality, such event has to be a subset of event A defined as follows:

A = the courier is of a distance at least $\frac{n}{2}$ from where
he picked up the package after $\lfloor n^{1.5} \rfloor$ steps

Again by triangular inequality, if event A happens, then the courier has to move at least $\frac{n}{4}$ along at least one coordinate. Without loss of generality, one may assume it is the y -coordinate. Then the displacement of the courier along y -coordinate can be written as the following i.i.d. sum:

$$Y_k = X_1 + X_2 + \cdots + X_k$$

where $P(X_i = 0) = \frac{1}{2}, P(X_i = \pm 1) = \frac{1}{4}$ with $E(X_i) = 0, \text{var}(X_i) = \frac{1}{2}$. So by Chebyshev Inequality,

$$P(A) \leq 2P\left(\left|Y_{\lfloor n^{1.5} \rfloor}\right| \geq \frac{n}{4}\right) \leq 2 \times \frac{n^{1.5}}{2n^2/16} = 16n^{-0.5} \ll 1$$

(2) Similar to Part (1), by triangular inequality, the event of interest has to be a subset of event A defined as follows:

A = the courier **has ever reached** a distance at least n from where
he picked up the package again $\lfloor n^{1.5} \rfloor$ steps

If A happens, then the courier has to have wandered at least $\frac{n}{2}$ along at least one coordinate. I.e.,

$$P(A) \leq 2P\left(\max_{k \leq n^{1.5}} |Y_k| \geq \frac{n}{2}\right)$$

Note that Y_k forms a martingale, by Doob's inequality,

$$P\left(\max_{k \leq n^{1.5}} |Y_k| \geq \frac{n}{2}\right) \leq \frac{E\left[\left|Y_{\lfloor n^{1.5} \rfloor}^2\right|\right]}{(n/2)^2} \leq \frac{n^{1.5}}{2(n/2)^2} = 2n^{-0.5} \ll 1$$

(3) Let $S_k \in \mathbb{Z}^2, k \geq 0$ be the location of courier at k th step. Define stopping times as follows:

$$\begin{aligned}\tau &= \min\{k \geq 0, S_k = 0\} \\ \hat{\tau} &= \min\{k \geq 0, |S_k| \geq 2^{n/4}\}\end{aligned}$$

Thus for the probability of interest we have,

$$P(\tau > 2^n) \leq P_{(n,0)}(\hat{\tau} > 2^n) + P_{(n,0)}(\hat{\tau} < \tau)$$

where $P_{(n,0)}$ stands for the distribution of simple random walk starting from $(n, 0)$. For the first term on RHS of the inequality above, by invariance principle, we have $S_{\lfloor 2^{n/2}t \rfloor} 2^{-\frac{n}{4}}, t \in [0, 1]$ converges to a 2D standard Brownian Motion B_t , where

$$P(|B_1| \geq 2) = c > 0$$

Thus for all sufficiently large n ,

$$\min_{|X_0| \leq 2^{n/4}} P_{X_0}(|S_{2^{n/2}}| \geq 2^{n/4}) \geq P_0(|S_{2^{n/2}}| \geq 2 \times 2^{n/4}) \geq P(|B_1| \geq 2) - \frac{c}{2} \geq \frac{c}{2}$$

Then we partite the 2^n into subintervals of length $2^{n/2}$. If the courier never level the neighborhood of radius $2^{n/4}$ within 2^n steps, he has to be so at time points $2^{n/2}, 2 \times 2^{n/2}, \dots, [2^{n/2}]2^{n/2}$ as wells. Thus by Markov property,

$$P_{(n,0)}(\hat{\tau} > 2^n) \leq \left[1 - \min_{|X_0| \leq 2^{n/4}} P_{X_0}(|S_{2^{n/2}}| \geq 2^{n/4})\right]^{[2^{n/2}]} \leq \left(1 - \frac{c}{2}\right)^{[2^{n/2}]} \ll 1$$

Now for the second term $p_2 = P_{(n,0)}(\hat{\tau} < \tau)$ we need to consider the potential kernel of $2D$ simple random walk, for any $x \in \mathbb{Z}^2$ define

$$a(x) = \sum_{k=0}^{+\infty} [P_0(S_k = 0) - P_0(S_k = x)]$$

By Kesten (1987, Lemma 1)¹, we have

- 1) $a(x)$ converges for all $x \in \mathbb{Z}^2$ and is nonnegative
- 2) For SRW S_k starting from any initial point, $a(S_{k \wedge \tau})$ is a nonnegative martingale
- 3) There is a constant C_0 such that

$$|a(x) - \log |x|/(2\pi) - C_0| = O(|x|^{-2})$$

Thus by martingale optional stopping theorem, we have

$$a((n, 0)) = (1 - p_2) \times 0 + p_2 E[a(S_{\hat{\tau}}) | \tau > \hat{\tau}]$$

Thus

$$p_2 = \frac{a((n, 0))}{E[a(S_{\hat{\tau}}) | \tau > \hat{\tau}]}$$

Recalling Condition 3 above,

$$a((n, 0)) \asymp \log(n)$$

$$E[a(S_{\hat{\tau}}) | \tau > \hat{\tau}] \asymp n$$

Thus we have $p_2 \ll 1$ and complete the proof.

¹H. Kesten, Hitting probabilities of random walks on $\mathbb{Z}^d$, Stochastic Process. Appl. 25 (1987) 165-184.

9 Questions

- (1) Show that there is no *periodic* sequence $a_1, a_2, a_3, \dots$ of signs $a_n \in \{\pm 1\}$ such that, for every *rational* number θ ,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

- (2) Show that there is no sequence $a_1, a_2, a_3, \dots$ of signs $a_n \in \{\pm 1\}$ such that, for every *rational* number θ

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq 2022?$$

- (3) Give an example of a sequence of signs $a_n \in \{\pm 1\}$ such that, for every $\theta \in \mathbb{Q} - \mathbb{Z}$,

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty,$$

and which takes each of the values $+1$ and -1 an infinite number of times.

9 Answer (1) To simplify the notation, we shall write $z := e^{2\pi i\theta}$. Suppose, by contradiction, that there is a sequence (a_n) , periodic with period $d \in \mathbb{N}$, such that

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n z^n \right| < +\infty,$$

whenever z is of the form $e^{2\pi i\theta}$ with $\theta \in \mathbb{Q}$.

For convenience, take N to be a multiple d , say $N = dM$. Since (a_n) is periodic, we can group the terms of the above sum as

$$\begin{aligned} \sum_{n=1}^{Md} a_n z^n &= (a_1 z + a_2 z^2 + \dots a_d z^d) + z^d (a_1 z + a_2 z^2 + \dots a_d z^d) \\ &\quad + \dots + z^{(M-1)d} (a_1 z + a_2 z^2 + \dots a_d z^d) \\ &= (a_1 z + a_2 z^2 + \dots a_d z^d) \left(1 + z^d + z^{2d} + \dots + z^{(M-1)d} \right). \end{aligned}$$

Take $\theta = k/d \in \mathbb{Q}$, for some $k \in \{0, \dots, d-1\}$ to be chosen later. Then $z^d = (e^{2\pi i k/d})^d = 1$, so

$$\sum_{n=1}^{Md} a_n z^n = M(a_1 z + a_2 z^2 + \dots a_d z^d).$$

To conclude the problem, it suffices to show that $a_1 z + a_2 z^2 + \dots a_d z^d$ is nonzero for some appropriate choice of k , since we would then have

$$\lim_{M \rightarrow +\infty} \left| \sum_{n=1}^{Md} a_n z^n \right| = \lim_{M \rightarrow +\infty} M |a_1 z + a_2 z^2 + \dots a_d z^d| = +\infty,$$

contradicting ??.

This is clear: if $a_1 z + a_2 z^2 + \dots a_d z^d$ vanished for all $k \in \{0, \dots, d-1\}$ (with $z = e^{2\pi i k/d}$), then the polynomial $P(X) = a_1 X + a_2 X^2 + \dots a_d X^d$ would have $d+1$ distinct complex roots

$$e^{2\pi i 0/d}, e^{2\pi i 1/d}, \dots, e^{2\pi i (d-1)/d} \quad \text{and} \quad 0,$$

which is impossible for a degree d polynomial.

(2) Suppose, by contradiction, that there is such a sequence (a_n) . At first, since

$$f_N := \sum_{n=1}^N a_n e^{2\pi i n \theta}$$

is a continuous function of θ , $|f_N(\theta)| \leq 2022$ for any $\theta \in \mathbb{R}$.

Now we calculate the L^2 norm of f_N . Notice that for $n \in \mathbb{Z}$,

$$\int_0^1 e^{2\pi i n \theta} d\theta = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0. \end{cases}$$

Then we have

$$\begin{aligned}
\int_0^1 |f_N|^2 d\theta &= \int_0^1 \sum_{n=1}^N a_n e^{2\pi i n \theta} \sum_{m=1}^N a_m e^{-2\pi i m \theta} d\theta \\
&= \sum_{n=1}^N \sum_{m=1}^N a_n a_m \underbrace{\int_0^1 e^{2\pi i (n-m)\theta} d\theta}_{\text{nonzero only if } n=m} \\
&= \sum_{n=1}^N a_n^2 = N
\end{aligned}$$

This shows that (a_n) cannot have the property that

$$\left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq 2022$$

for all N and $\theta \in \mathbb{R}$! If this were true, we would have $|f_N(\theta)| \leq 2022$ for all N and $\theta \in \mathbb{R}$. $\int_0^1 |f_N|^2 d\theta \leq 2022^2$, contradicting $\int_0^1 |f_N|^2 d\theta = N$ if $N > 2022^2$.

(3) (We remark that eventually-constant sequences satisfy the inequality by the geometric series formula, but are explicitly discarded in the statement of the problem.)

To build up some intuition for the problem, let us take $\theta = 1/3$ and $N = 3M$. We want to find a sequence a_i such that

$$\sup_{M \in \mathbb{N}} \left| \left(\sum_{n=0}^{M-1} a_{3n+1} \right) \cdot 1 + \left(\sum_{n=0}^{M-1} a_{3n+2} \right) \cdot e^{2\pi i/3} + \left(\sum_{n=0}^{M-1} a_{3n+3} \right) \cdot e^{4\pi i/3} \right| < +\infty$$

Geometrically, this corresponds to scaling the three vectors 1 , $e^{2\pi i/3}$ and $e^{4\pi i/3}$, and computing the norm of the sum of these scaled vectors.

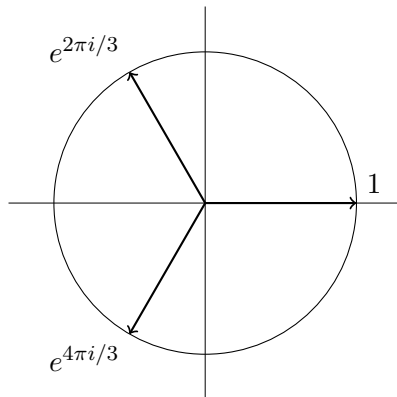


Figure 3: Case $\theta = 1/3$.

Since $1 + e^{2\pi i/3} + e^{4\pi i/3} = 0$, we should try to scale these three vectors by almost the same amount in order to obtain good cancellation. In other words, the distribution of the number of 1's and -1 's in the sequence $(a_{3n+i})_{n \geq 1}$ should be approximately the same for $i = 1$, $i = 2$ and $i = 3$. Of course, a similar property should hold for any natural number (greater than 1 since $\theta \notin \mathbb{Z}$) in place of 3.

There is a variety of ways to construct such a sequence (a_n) . One way to do it is as follows:

$$\begin{array}{c|c|c|c|c|c|c|c|c|c|c|c|c} n & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \hline a_n & +1 & -1 & -1 & +1 & +1 & +1 & +1 & +1 & +1 & -1 & -1 \end{array},$$

where $+1$ appears $1!$ time, then -1 appears $2!$ times, then $+1$ appears $3! = 6$ times, and so on. Let us now prove that this sequence has the desired property.

For $k \geq 0$, let $f(k) = 0! + 1! + \dots + k!$, and let $I_k \subset \mathbb{N}$ be the set of integers n with $f(k) \leq n < f(k+1)$. The sequence above is that which takes the value $a_n = (-1)^k$ for $n \in I_k$. The key to our construction is that, for every integer $q \geq 2$, the length of I_k , namely $(k+1)!$, is divisible by q whenever k is sufficiently large.

Let $\theta = p/q$ be a rational number, with $\gcd(p, q) = 1$ and $q > 1$. Then

$$\sum_{n=1}^N a_n e^{2\pi i n \theta} = \sum_{k \geq 0} (-1)^k \sum_{\substack{n \in I_k \\ n \leq N}} e^{2\pi i n p/q}.$$

We need to show that this expression is bounded above by a constant that does not depend on N .

Consider, for each fixed $k \geq 0$, the inner sum $S_k := \sum_{\substack{n \in I_k \\ n \leq N}} e^{2\pi i n p/q}$.

- (i) If $k < q$, we will be able to use the trivial bound $|S_k| \leq f(k+1) - f(k) = (k+1)!$ (obtained by the triangle inequality). This will be enough since there are only a finite number of such k .
- (ii) If $k \geq q$, we will exploit the particular way in which we constructed the sequence (a_n) to obtain sufficient cancellation.
 - (a) If $k \geq q$ and $f(k+1) \leq N$, we have perfect cancellation, i.e. $S_k = 0$. Indeed, in this case,

$$S_k = \sum_{n=f(k)}^{f(k+1)-1} e^{2\pi i n p/q} = \frac{(e^{2\pi i p/q})^{f(k+1)} - (e^{2\pi i p/q})^{f(k)}}{e^{2\pi i p/q} - 1}$$

by the geometric series formula. Since $k \geq q$, $(k+1)!$ is divisible by q , so the numerator vanishes:

$$(e^{2\pi i p/q})^{f(k)+(k+1)!} - (e^{2\pi i p/q})^{f(k)} = (e^{2\pi i p/q})^{f(k)} (e^{2\pi i p(k+1)!/q} - 1) = 0.$$

- (b) If $k \geq q$ and $N \in I_k$ (this happens for exactly one value of k), we have

$$S_k = \sum_{n=f(k)}^N e^{2\pi i n p/q} = \frac{(e^{2\pi i p/q})^N - (e^{2\pi i p/q})^{f(k)}}{e^{2\pi i p/q} - 1}.$$

By the triangle inequality, we deduce the bound

$$|S_k| \leq \frac{\left| (e^{2\pi ip/q})^N \right| + \left| (e^{2\pi ip/q})^{f(k)} \right|}{|e^{2\pi ip/q} - 1|} = \frac{2}{|e^{2\pi ip/q} - 1|}$$

which is independent of N .

(c) If $k \geq q$ and $N < f(k)$, we have of course $S_k = 0$ since the sum is empty.

Therefore,

$$\left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| \leq \sum_{k=0}^{q-1} |S_k| + \sum_{k \geq q} |S_k| \leq f(q) + \frac{2}{|e^{2\pi ip/q} - 1|}.$$

This shows that

$$\sup_{N \in \mathbb{N}} \left| \sum_{n=1}^N a_n e^{2\pi i n \theta} \right| < +\infty.$$

10 Program Design For The Opening Ceremony

Suppose you are chosen as a technology assistant by the director of the opening ceremony for the 2022 Winter Olympics, and your job is to evaluate the program proposals. One of the backup programs is a skating show of a ensemble of drones dressed as mascots, which are moving along a circle. Since the number of the drones is sufficiently large, we can use a probability density function $\rho(t, v) (\geq 0)$ to represent the distribution of the drones. Here, $v \in \mathbb{R}$ is the line speed, and for a given time t , and two speeds $v_1 < v_2$,

$$\int_{v_1}^{v_2} \rho(t, v) dv$$

is the probability of find a drone with its speed between v_1 and v_2 .

Suppose that the dynamics of the density function is governed by

$$\rho_t + ((u(t) - v)\rho)_v = \rho_{vv}, \quad v \in \mathbb{R}, \quad t > 0,$$

where $u(t)$ is the command speed.

(1) To investigate proper choices of the command speed, D.B. propose that we shall choose

$$u(t) = u_0 + u_1 N(t)$$

where $u_0 > 0$, $u_1 > 0$ and $N(t)$ is the average of the positive part of the speed $v_+ = \max\{0, v\}$, i.e.,

$$N(t) = \int_{-\infty}^{+\infty} v_+ \rho(t, v) dv = \int_0^{+\infty} v \rho(t, v) dv.$$

But you claim that if $u_1 > 1$, $N(t)$ may become unbounded in evolution, such that the drones may be out of order. Can you prove it? (For simplicity, the contributions of ρ and its derivatives at $|v| \rightarrow +\infty$ are neglected.)

(2) After taking those advices, the directly is wondering whether the drones will be evenly distributed along the circle. Thus, we shall consider the joint density function $p(t, x, v) (\geq 0)$ of position and speed, where $x \in [0, 2\pi]$ is the position coordinate on the circle. Clearly, $\int_0^{2\pi} p(t, x, v) dx = \rho(t, v)$. Suppose that the governing equation for $p(t, x, v)$ is

$$p_t + vp_x + ((u(t) - v)p)_v = p_{vv}, \quad x \in [0, 2\pi], \quad v \in \mathbb{R}, \quad t > 0.$$

Because, the drones are circulating around, the following boundary condition is satisfied

$$p(t, 0, v) = p(t, 2\pi, v), \quad v \in \mathbb{R}, \quad t > 0.$$

You have a feeling that, no matter how the drones are distributed initially, they will become almost evenly distributed very quickly. Can you prove or disprove this statement? (For simplicity, the contributions of p and its derivatives at $|v| \rightarrow +\infty$ are neglected.)

10 Answer (1) We define the average speed

$$M(t) = \int_{-\infty}^{+\infty} v\rho(t, v) dv.$$

By direct calculations, we find

$$\begin{aligned} \frac{d}{dt}M(t) &= \frac{d}{dt} \int_{-\infty}^{+\infty} v\rho(t, v) dv \\ &= \int_{-\infty}^{+\infty} v\rho_t(t, v) dv \\ &= \int_{-\infty}^{+\infty} v(-(u(t) - v)\rho + \rho_v)_v dv \\ &= \int_{-\infty}^{+\infty} ((u(t) - v)\rho - \rho_v) dv \\ &= u(t) - M(t) \\ &= u_0 + u_1N(t) - M(t). \end{aligned}$$

Because $M(t) \leq N(t)$ and clearly $N(t) \geq 0$, we get

$$\frac{d}{dt}M(t) \geq u_0 + (u_1 - 1)N(t) \geq u_0 > 0.$$

Then we conclude that when $t \rightarrow +\infty$, $M(t) \rightarrow +\infty$.

And since $N(t) \geq M(t)$, $N(t)$ also diverges to $+\infty$.

(2) Due to the periodic boundary condition in x , we can write the solution in the form of Fourier series:

$$p(t, x, v) = \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} p_k(t, v)e^{ikx}.$$

where

$$p_k(t, v) = \int_0^{2\pi} p(t, x, v)e^{-ikx} dx.$$

And we aim to show for $k \neq 0$, $p_k(t, v)$ shall decay.

Due to the orthogonality, we find p_k satisfies

$$\partial_t p_k + ikvp_k = -\partial_v((u(t) - v)p_k)_v + \partial_{vv}p_k, \quad t > 0, \quad k \in \mathbb{Z}, \quad v \in \mathbb{R}.$$

Next we apply the Fourier transform in v . Let

$$\hat{p}_k(t, \xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} p_k(t, v)e^{-iv\xi} dv,$$

and we can derive that $\hat{p}_k(t, \xi)$ satisfies

$$\partial_t \hat{p}_k + (\xi - k)\partial_\xi \hat{p}_k = -(i\xi u(t) + \xi^2) \hat{p}_k.$$

Next we use the method of characteristics. Consider the characteristic equation

$$\frac{d}{dt}\xi_k(t) = \xi_k(t) - k,$$

whose solution is given by

$$\xi_k(t) - k = e^{t-s} (\xi_k(s) - k).$$

Along the characteristics, we have

$$\hat{p}_k(t, \xi_k(t)) = \hat{p}_k(0, \xi_k(0)) e^{-\int_0^t (\xi_k(s))^2 + iu(s)\xi_k(s) ds}$$

which leads to

$$\hat{p}_k(t, \xi) = \hat{p}_k(0, (\xi - k)e^{-t} + k) e^{-H_{t,k}(\xi - k)},$$

where

$$H_{t,k}(z) = \frac{1 - e^{-2t}}{2} z^2 + 2k(1 - e^{-t})z + i \int_0^t u(s) e^{-(t-s)} ds z + k^2 t + ik \int_0^t u(s) ds.$$

Applying the inverse Fourier transform, we get

$$p_k(t, v) = e^{ikv} (p_{t,k} * G_{t,k})(v).$$

where

$$p_{t,k}(y) := e^t p_{0,k}(e^t y), \quad p_{0,k}(v) := e^{-ikv} \int_0^{2\pi} p_{\text{init}}(x, v) e^{-ikx} dx.$$

and $G_{t,k}(z)$ is a complex-valued Gaussian function defined as follows

$$G_{t,k}(z) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(z - \mu)^2}{2\sigma^2}\right) \exp(ik\Theta) \exp(-k^2 D).$$

$$\mu(t) = \int_0^t e^{-(t-s)} u(s) ds, \quad \sigma(t) = \sqrt{1 - e^{-2t}},$$

$$\Theta(t, z) = -\frac{2(z - \mu(t))}{1 - e^{-t}} - \int_0^t u(s) ds, \quad D(t) = t - 2\frac{1 - e^{-t}}{1 + e^{-t}}.$$

We observe that when $t > 0$, $G_{t,k}$ decays due to the factor $\exp(-k^2 D)$, and for t sufficiently large, it decays exponentially fast.

At last, with the convolution inequality, we get

$$\begin{aligned} \|p_k(t, v)\|_{L^1(\mathbb{R})} &= \|e^{ikv} (p_{t,k} * G_{t,k})(v)\|_{L^1(\mathbb{R})} \leq \|p_{t,k}(g)\|_{L^1(\mathbb{R})} \|G_{t,k}\|_{L^1(\mathbb{R})} \\ &= \|p_{0,k}(g)\|_{L^1(\mathbb{R})} e^{-k^2 D(t)}. \end{aligned}$$

This shows that for $k \neq 0$, $p_k(t, v)$ decays, and $p(t, x, v)$ is approaching $\frac{1}{\pi} p_0(t, v)$, which is an even distribution in space.